after the initiation of propyl-thiouracil treatment and thyroidectomy, respectively. Serum luteinizing hormone (LH) levels significantly reduced in both groups on the day of diestrus-1 about 1 month later. Hypothyroid rat shows the high progesterone and low testosterone levels. No significant changes in inhibin and estradiol levels were detected. The serum levels of FSH decreased in the thyroidectomy group. The irregular estrous cycle and the above changes in hormone levels were recovered by administration of T4. Compensatory secretions of FSH and LH induced by ovariectomy were enhanced by thyroidectomy and suppressed by T4 treatment. These results suggest that thyroid hormones play a role in the regulation of reproductive hormones secretion in the cyclic ovary."[98]

Whenever abnormal or dysregulated FSH and LH patterns are recognized on a female hormone profile, and especially when progesterone is elevated with depressed LH not related to exogenous progesterone, thyroid dysfunction needs to be evaluated. Additionally, many times when hypothyroidism is present, progesterone resistance will result, causing the progesterone receptor sites to lose their responsiveness to progesterone and progesterone levels will elevate. These individuals will present with a complex set of symptoms that may be specific to their menstrual disorders and not their thyroid, so it is necessary for the culprit, the thyroid, to be evaluated.

Male Testicular Cell Activity and the Thyroid

"Leydig cell steroidogenic activity under basal and stimulated conditions was studied in hypothyroid rats." In a study, "hypothyroidism was induced at a prepubertal age (30 days postpartum) by surgical thyroidectomy, and L-thyroxine (T4) supplementation (6 micrograms/100 g body weight/day for 30 days) to hypothyroid rats was begun after 30 days. Hypothyroidism for 60 days reduced serum LH and FSH without affecting prolactin. Serum and intratesticular testosterone and the specific activity of Leydig cell 3 beta- and 17 beta-hydroxysteroid dehydrogenases diminished in hypothyroid rats. The stimulatory effect of LH on Leydig cell steroidogenic activity and cAMP was also adversely affected in hypothyroid rats. All these changes were reversed by T4 supplementation. The present results of the study suggest that prepubertal hypothyroidism suppresses both basal and stimulated Leydig cell activity in adult rats."[99]

"Thyroid hormones are important for the growth and development of many tissues. Altered thyroid hormone status causes testicular abnormalities. For instance, juvenile hypothyroidism/neonatal transient hypothyroidism induces macroorchidism, increases testicular cell number (Sertoli, Leydig, and germ cells) and daily sperm production. Triiodothyronine (T3) receptors have been identified in sperm, developing germ cells, Sertoli, Leydig, and peritubular cells. T3 stimulates Sertoli cell lactate secretion as well as mRNA expression of inhibin-alpha, androgen receptor, IGF-I, and IGFBP-4. It also inhibits Sertoli cell mRNA expression of Müllerian inhibiting substance (MIS), aromatase, estradiol receptor, and androgen binding protein (ABP) and ABP secretion. T3 directly increases Leydig cell LH receptor numbers and mRNA levels of steroidogenic enzymes and steroidogenic acute regulatory protein. It stimulates basal and LH-induced secretion of progesterone, testosterone, and estradiol by Leydig cells. Steroidogenic factor-1 acts as a mediator for T3-induced Leydig cell steroidogenesis. "It is clear that T3 directly regulates Sertoli and Leydig cell functions."[100]

Although hypothyroidism is not common in men, it does occur and it typically leads to severe alterations in male hormone profiles. The impacts of thyroid hormones on male androgenic pathways are so diverse that potentially any abnormal pattern could be secondary to thyroid dysfunction. These patterns become difficult to evaluate as a thyroid problem since the male's dysfunctional androgen disorders are the presenting symptomatology. Thyroid hormones can directly cause abnormal male hormone imbalances. On the other hand, when Hashimoto's is identified in a male, it is common to see antibodies against the testicular cells, testosterone, and/or sperm. Therefore, the connection with abnormal male hormone patterns and thyroid can be directly related to abnormal thyroid hormone influences, or from a polyendocrine autoimmune disorder that is disrupting both thyroid and testicular function.

Gonadal Steroidogenesis and the Thyroid

Hypothyroidism in men has been shown to cause diminished libido, impotence, and oligospermia. It appears that hypothyroidism alters the metabolism of both androgens and estrogens. Although hypothyroidism is not as common in males, it must be ruled out with altered states of male hormone

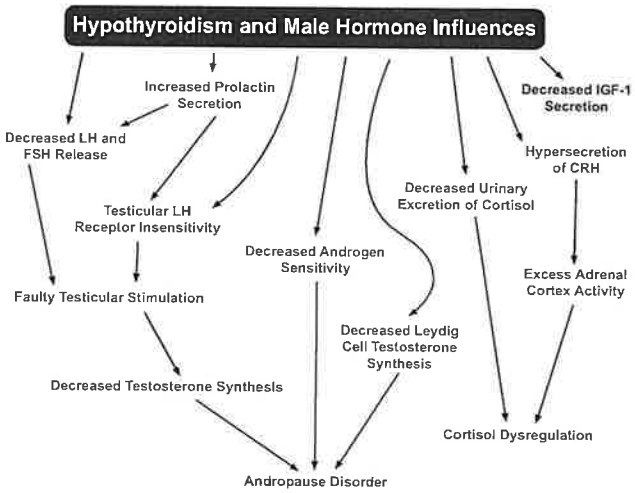
steroidogenesis.[101] "Inconsistent alterations in gonadal steroidogenesis and pituitary functions have been reported in hypothyroid males." A study "compared the lipid and endocrine profiles of the euthyroid and hypothyroid [thyroid-stimulating hormone (TSH) >100 mIU l(-1)] males. Hypothyroidism was found to be associated with an increase in the circulating level of total cholesterol and LDL-cholesterol (LDL-C) and a reduction in the levels of progesterone and testosterone, without any change in the serum levels of estradiol and gonadotropins. The failure of gonadotropins to rise could be accounted by a normal level of serum estradiol in the hypothyroid male. A mild hyperprolactinemia was also noted in the hypothyroid patients. The reduction in serum testosterone level could be explained by (i) a reduced uptake of LDL-C by the Leydig cells and thereby a reduction in the synthesis of progesterone and consequentially testosterone, (ii) a further reduction in the rate of conversion of progesterone to testosterone, (iii) a higher rate of conversion of testosterone to estradiol, (iv) a decrease in serum triiodothyronine and (v) hyperprolactinemia." The study concluded that a "rise in TSH needs to be investigated as a cause of the suppression of gonadal steroidogenesis."[102]

"The influence of hypothyroidism on testicular steroidogenesis was investigated by evaluating the production of testosterone and its precursors by isolated testes from adult male rats. Animals were made hypothyroid starting from the 4th week of life either by daily oral administration of 0.1% methimazole (MMI) or by surgical thyroidectomy (TZ). Half of the thyroidectomized rats were i.p. injected with 3 gamma T3/100 g body weight on alternate days during the last three weeks before sacrifice. Hypothyroidism is associated with a severe retardation of body growth, which appears more marked in thyroidectomized than in MMI treated rats; no significant variations in testis weight are observed. Administration of T3 does not completely restore body weight. A significant decrease in the "in vitro" production of testosterone and its precursors by testes isolated from hypothyroid rats is observed. This effect is more evident in thyroidectomized rats where a marked drop in the "in vitro" production of some testosterone delta 4 precursors is associated with the increase in DHEA/delta 4 ratio. T3 injection to thyroidectomized rats only partially restores the "in vitro" testosterone production. Results suggest that as the degree of hypothyroidism became more severe, the rate of testosterone production decreases and testicular steroidogenesis changes from the delta 4 to delta 5 metabolic pathway as a consequence of the impairment of 3-beta-ol-dehydrogenase activity."[103]

"Thyroid hormone deficiency affects all tissues of the body, including multiple endocrine changes that alter growth hormone, corticotropin, glucocorticoids, and gonadal function. Primary hypothyroidism is associated with hypogonadotropic hypogonadism, which is reversible with thyroid hormone replacement therapy. In male children, follicle-stimulating hormone (FSH) is elevated and associated with testicular enlargement without virilization. Men with primary hypothyroidism have subnormal responses of luteinizing hormone (LH) to gonadotropin-releasing hormone (GnRH) administration and normal response to human chorionic gonadotropin (hCG). Free testosterone concentrations are reduced in men with primary hypothyroidism and thyroid hormone replacement normalizes free testosterone concentrations. In men with primary hypothyroidism, prolactin is not consistently elevated (except in men and children with longstanding severe primary hypothyroidism), but prolactin declines following thyroid hormone replacement therapy. Thyroid hormone is known to affect sex hormone-binding hormonal globulin (SHBG) concentrations. Men with hyperthyroidism have elevated concentrations of testosterone and SHBG. Thyroid hormone therapy in normal men may also duplicate this elevation. In addition, estradiol elevations are observed in men with hyperthyroidism, and gynecomastia is common in them as well. In contrast to patients with primary hypothyroidism, men with hyperthyroidism exhibit hyperresponsiveness of LH to GnRH administration and subnormal responses to hCG. Radioactive iodine therapy (RAI) of men treated for thyroid cancer produces a dose-dependent impairment of spermatogenesis and elevation of FSH up to approximately 2 years. Permanent testicular germ cell damage may occur in men treated with high doses of RAI. RAI commonly increases serum concentrations of FSH and LH while reducing inhibin B levels without affecting serum concentrations of testosterone. Thus, radioiodine therapy transiently impairs both germinal and Leydig cell function that usually recover by 18 months post therapy."[104]

Although hypothyroidism is not common in men, it does occur and it typically leads to severe alterations in male hormone profiles. The impacts of thyroid hormones on male androgenic pathways are so diverse that

potentially any abnormal pattern could be secondary to thyroid dysfunction. These patterns become difficult to evaluate as a thyroid problem since the male's dysfunctional androgen disorders are the presenting symptomatology. Thyroid hormones can directly cause abnormal male hormone imbalances. On the other hand, when Hashimoto's is identified in a male, it is common to see antibodies against the testicular cells, testosterone, and/or sperm. Therefore, the connection with abnormal male hormone patterns and thyroid can be directly related to abnormal thyroid hormone influences, or from a polyendocrine autoimmune disorder that is disrupting both thyroid and testicular function.



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Amenorrhea and the Thyroid

"The effect of thyroxine treatment in cases of functional amenorrhea accompanied by low thyroid function was studied. During six months of follow-up, regular or single menstrual cycles were restored in 10 of the 17 patients with thyroxine treatment and in four of the 11 patients without thyroxine treatment. During thyroxine treatment, thyroxine and free thyroxine index levels were elevated and serum testosterone levels decreased, but no significant changes were observed in serum estradiol, prolactin, follicle-stimulating hormone, or luteinizing hormone levels. In the thyroid-releasing hormone/luteinizing-releasing hormone test, the thyroid-stimulating hormone response was markedly suppressed, but there were no changes in follicle-stimulating hormone, luteinizing hormone, or prolactin responses. In one patient with primary hypothyroidism, regular menstrual cycles were restored and serum prolactin and thyroid-stimulating hormone levels normalized during thyroxine treatment. Thyroxine treatment seems to have benefits in some amenorrheic patients with low normal thyroid function."[105]

Clinically, when individuals have hypothyroidism, a common feature is an increase in prolactin, which causes suppression of LH release, loss of progesterone receptor-site sensitivity, and follicle FSH sensitivity, leading both to failure to have proper ovulation as well as pituitary discoordination. The use of birth control pills or natural hormones in these cases will be devastating to the physiological feedback loop. Additionally, the use of herbs to stimulate the ovarian or reproductive glandular system will prove to be ineffective. Careful analysis and support of thyroid function should be determined with amenorrhea caused by thyroid disturbances.

LH, FSH, and the Thyroid

In one study, "The effects of hypothyroidism and hyperthyroidism on serum LH and FSH were determined in both intact and castrated male and female rats. Thyro-parathyroidectomy (TX) for 30 days in rats with intact gonads resulted in a significant reduction in serum LH and FSH and also in a decrease in serum testosterone in males. Administration of 2.5 mug thyroxine (T4)/100 g BW to TX rats of both sexes returned serum LH and FSH levels to those of intact rats, and in males also restored normal serum testosterone levels. TX superimposed upon castration resulted in a significantly greater increase in serum LH and FSH than produced by castration alone. Administration of 2.5 mug T4/100 g body weight to castrated-TX rats reduced serum LH and FSH values to those of castrated rats whereas 10 mug T4/100 g BW evoked a further decrease in serum LH but no additional reduction in serum FSH. When both 2.5 mug T4/100 g BW and 2 mug estradiol benzoate were injected into TX-ovariectomized rats, the decrease in serum LH and FSH was much greater than produced by T4 alone. These observations indicate that hypothyroidism results in decreased release of LH and FSH in rats with intact gonads, and in increased release of LH and FSH in castrated rats of both sexes. Administration of a replacement dose of T4 can restore LH and FSH release to normal in TX rats with intact gonads, and to castration levels in TX-castrated rats."[106]

Research has "demonstrated that the thyroid hormone

synergizes with follicle-stimulating hormone (FSH) to exert stimulatory effects on granulosa cell differentiation and function, suggesting that it plays a physiological role in amplifying FSH-mediated differentiation of granulosa cells. The adequate differentiation of these cells, followed by normal follicle development, is indispensable for ovulation and subsequent corpus luteum formation." A study was conducted to evaluate "the clinical implications of the thyroid hormone in the induction of ovulation and corpus luteum function." The results demonstrated that "serum levels of total 3,5,3'-triiodothyronine (T3) and total thyroxine (T4) as well as free T3 and T4 were significantly lower in patients with weight loss amenorrhea compared to normal cycling women."[107]

"In an effort to determine the influence of hypothyroidism on the hypothalamopituitary-ovarian axis plasma levels of luteinizing hormone (LH), follicle-stimulating hormone (FSH), estrogen and progesterone were measured daily for 10 to 14 consecutive days in five women with hypothyroidism. In all subjects investigated, the levels of all the hormones measured were low in comparison with the values in euthyroid women. FSH/LH ratios were, however, higher in hypothyroid women than in both phases of the menstrual cycle in euthyroid controls. It is postulated that changes in the FSH/LH ratio may be related to the low levels of estrogen and probably to failure of ovulation in hypothyroid women."[108]

Hypothyroidism may lead to altered FSH levels, which may lead to immature follicles and subsequent infertility. Suppressed LH levels will typically lead to altered ovulation in timing or abnormal luteal phase distribution of progesterone. These changes may present as miscarriages, depression in the second half of the cycle, migraines at the second half of the cycle, or a history of miscarriages.

Prolactin, CRH, and the Thyroid

A study evaluating males indicated that hypothyroidism causes adrenal dysfunction directly and results in hypersecretion of corticotropin-releasing hormone (CRH) and arginine vasopressin (AVP) from the hypothalamus. "The pituitary response to luteinizing hormone-releasing hormone (LHRH) was lowered, whereas the testicular response to hCG was not changed in hypothyroid rats. Hypothyroidism suppressed copulatory behavior in male rats. These results suggest that hypothyroidism probably causes dysfunction in [the] gonadal axis at the hypothalamic-pituitary level in male rats. In adult female rats, hypothyroidism inhibited the follicular development accompanied estradiol secretion, whereas plasma concentrations of progesterone and prolactin (PRL) increased in hypothyroid female rats. Hypothyroidism significantly increased the pituitary content of vasoactive intestinal peptide (VIP) though it did not affect dopamine synthesis. These results suggest that hypothyroidism increases pituitary content of VIP, and this increased level of VIP likely affects PRL secretion in a paracrine or autocrine manner. In female rats, inhibition of gonadal function in hypothyroid subjects was mediated by hyperprolactinemia in addition to hypersecretion of endogenous CRH."[109]

A study evaluated "25 women aged 22 to 26 years with a diagnosis of prolactinoma moderately increased in primary sterility. Among 25 women under observation, 7 (28%) showed a hormone picture of subclinical hypothyroidism. In fact the evaluation of TSH after stimulation with TRH showed a curve typical of the subclinical hypothyroidism. In 8 women (32%) the authors have found an increase of the thyroid gland, and in 3 patients (12%) galactorrhea. The increase of volume of the thyroid gland was constant in all the women with subclinical hypothyroidism. Considering the results obtained by this study, the authors point out a correlation between subclinical hypothyroidism-hyperprolactinemia and sterility."[110]

Infertility and the Thyroid

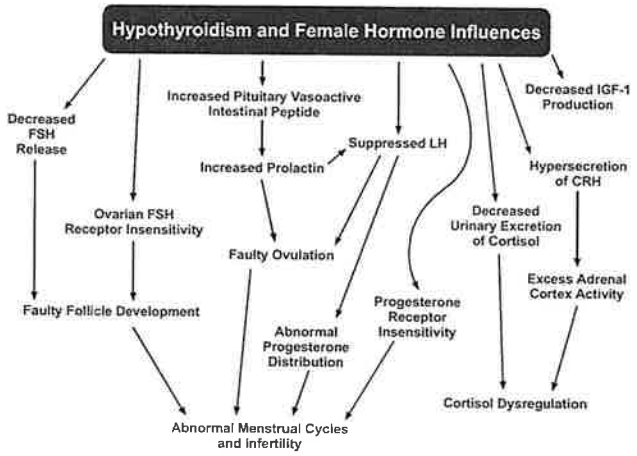
"Female hypothyroid (hyt) mice are infertile, but the reason for this infertility is not yet known. The present study was conducted to determine whether hypothyroidism induced infertility in immature and mature hyt mice. Furthermore, animals were treated with thyroxine and gonadotrophins at different times to determine whether infertility was due to failure of follicular development, implantation, or pregnancy. There were no significant differences in the numbers of ovulated eggs induced by gonadotrophin treatment or the percentages of eggs developed in vitro among immature normal controls, hyt, and thyroxine-treated hyt mice. Mature hyt mice showed continuous diestrus, and ovulated significantly fewer eggs after gonadotrophin treatment and failed to establish pregnancy after mating compared with mature control mice. Mature hyt mice had significantly fewer corpora lutea > 500 microhm in diameter and significantly lower progesterone concentrations. Thyroxine treatment before mating in hyt mice resulted in well-developed corpora lutea,

an increase in progesterone and normal pregnancy, regardless of subsequent thyroxine administration. In conclusion, infertility occurs in mature rather than immature hyt mice, is due to the failure of follicular development and pregnancy, and can be reversed by thyroxine treatment before mating."[111]

"For sensitive assessment of thyroid function a TRH stimulation test using 200 micrograms TRH i.v. was routinely performed in 304 women admitted for evaluation and treatment of infertility. In 37 cases (12.2%) the reaction of TSH 30 minutes after injection of TRH i.v. was enhanced (by definition of a peak TSH level greater than 25 mIU/l), according to mild or subclinical hypothyroidism. Approximately 14 (14/37 = 37.8%) of these patients were found to have slightly elevated serum PRL values (mean PRL greater than 15 ng/ml). Cycle analysis by means of basal body temperature and evaluation of progesterone and estradiol values supplied evidence of luteal phase deficiency in 8 and anovulation in 3 cases. Another group of 11 patients with hypothyroidism involved oligo-/amenorrhea, hirsutism and hyperandrogenaemia. After treatment with 50-150 micrograms l-thyroxine daily for at least 4 to 6 weeks, elevated PRL values significantly decreased (mean level less than 15 ng/ml, p less than 0.01) in 9 out of 12 patients and testosterone levels slightly decreased in 5 out of 8 patients. An improvement of the cyclical ovarian function could be observed by the significant increase of the average progesterone concentration in the luteal phase. During therapy with l-thyroxine, 4 pregnancies occurred. From these results we conclude that mild hypothyroidism may cause ovarian insufficiency. Assessment of thyroid function should be mandatory in infertile patients with elevated prolactin levels or chronic anovulation."[112]

Pituitary-Ovarian Axis and the Thyroid

"The pituitary-ovarian axis was studied after withdrawal of thyroid hormone in 131 I-radiothyroidectomized adult female rats. Estrous cycles became prolonged and irregular within 2 weeks after the supply of thyroid hormone was stopped. If an LH surge occurred in hypothyroid rats on the day of vaginal pro-oestrus it was significantly greater in rats which had been made hypothyroid for 4-5 weeks than in controls; in hypothyroid rats with an LH surge on pro-oestrus, plasma progesterone showed a rise similar to that in controls at pro-oestrus; the ovulation rate was decreased in hypothyroid rats. About half of the rats from which blood was sampled daily in the afternoon between 7 and 18 days after triiodothyronine (T3) withdrawal had 1 day of pro-oestrus; on this day the LH surge was higher than in controls. On days 2 and 1 before and days 1 and 2 after this pro-oestrus, plasma progesterone was similar to that of controls on days 2 and 1 before and days 1 and 2 after pro-oestrus respectively. However, progesterone was higher in the period before and after these days. The other hypothyroid rats showed no pro-oestrus and no LH surge during this period, while their plasma progesterone levels were high on all days. On the morning of day 10 after T3 withdrawal and 5 days after the preceding pro-oestrus, most hypothyroid rats had high progesterone and low oestradiol plasma levels."[113]



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Low Thyroid Status

↓

Reduced Sex Hormone-Binding Levels

↓

Increased Free-Fraction Hormones
- **Estrogen**
- **Testosterone**

↓

Receptor-Site Resistance

↓

Symptoms of Reproductive Hormone Imbalance

Glucose Metabolism and the Thyroid

"The effect of different thyroid states on glucose homeostasis was investigated during metabolic adaptation to starvation in the conscious unrestrained miniature pig. Moderate hyperthyroidism increased the rate of glucose turnover, whereas hypothyroidism was without effect. Glucose recycling was elevated in hyperthyroid pigs, and reduced after thyroidectomy. Supplementary doses of T4 normalized total glucose recycling. Glucose metabolic clearance rate and pool size were unaffected by thyroid hormones. During starvation, serum insulin showed a similar decrease in all thyroid states; glucagon increased in euthyroid and hypothyroid pigs, although it was already elevated in the hyperthyroid fed state. Serum cortisol levels, although varying, were enhanced in hyperthyroid and hypothyroid-T4-treated pigs. Glucogenic precursor concentration and cumulative urinary N-excretion were increased in hyperthyroid pigs. It is concluded that 1) even a moderate hyperthyroidism produces an increase in glucose turnover and a concomitant acceleration in protein breakdown, and 2) thyroid hormone is essential for the starvation-induced total glucose recycling."[114]

Estrogen Metabolites and the Thyroid

As estrogens are metabolized in the phase I reactions into secondary metabolites, thyroid hormones may influence the activity of cytochrome P450 1A1 and 1A2. As estrogens are hydroxylated in phase I, they may have a hydroxyl group attached to the 2-carbon, the 4-carbon, or the 16-alpha-carbon. The 4- and 16-alpha-carbons produce 4- or 16-alpha hydroxyestrone or estradiol and hydroxylation to the 2-carbon yields 2-hydroxyestrone or estradiol. The 2-hydroxy metabolites are considered protective for estrogen proliferation disorders, especially breast cancer. The 4- and 16-carbon metabolites are considered proliferative and have been shown to increase the risk of breast cancer. In states of hypothyroidism, the metabolism of estrogens is shifted in such a way as to favor the 16-alpha-hydroxylation over the 2-hydroxylation. Therefore, hypothyroidism decreased the 2:16 estrogen metabolite ratio in such a way as to increase the risk for estrogen-related proliferation.[115]

> *Estrogen Metabolic Dysfunction*
> For the treatment of this condition:
> Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- Estrogen levels

The use of indole-3-carbinol and selective estrogen-receptor nutritional support, should always be considered in women, but they definitely should be considered to support thyroid function and estrogen levels.

Hypothyroidism and Estrogen Metabolite Metabolism

Hypothyroidism

↓

Diminished thyroid hormone influence on P450 1A1 and 1A2 cytochrome P450 enzymes

↓

Decreased 2:16 estrogen/metabolite ratio

↓

Promoted estrogenic tissue proliferation

> For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Progesterone-Thyroid Interplay

There are interactive influences of progesterone on thyroid metabolism and there are interactive influences of thyroid hormones on progesterone receptor-site expression. Progesterone has the potential to influence the metabolic expression of thyroid peroxidase (TPO) activity, which is the rate-limiting enzyme in the thyroid for thyroxine synthesis. It is this influence that is partially responsible for the increased body temperature women may express during ovulation. Ovulation takes place from an LH surge and therefore causes an immediate surge of progesterone. This upregulates the activity of TPO and increases the fluctuation of thyroid hormones. It is these fluctuations of thyroid hormones that increase metabolic activity and, therefore, body temperature. Clinically, one observes patients who suffer from subclinical thyroid insufficiency symptoms and concomitant symptoms of progesterone insufficiency. These patients do not typically have abnormal thyroid profiles, but the T4 levels are on the lower end of the normal range with a normal TSH. Their progesterone insufficiency symptoms include heavy menstrual cycles, an inability to lose weight, mid-cycle/luteal phase depression, headaches, etc. The management of their progesterone imbalance generally resolves their symptoms of thyroid insufficiency. With menstruating women who have patterns of progesterone insufficiency, the use of exogenous progesterone is not recommended. When menstruating women have symptoms of progesterone insufficiency, it is rarely because their bodies have lost the ability to produce progesterone, but rather a disorganized ovarian-pituitary feedback loop. This usually occurs when the LH surge is compromised from chronic adrenal axis overactivity, pituitary suppression from post-contraceptive use, or pituitary suppression from postpregnancy. It is important to normalize progesterone levels in these cases.

In menopausal women, exogenous progesterone may be appropriate to improve metabolism. Unlike menstruating women, menopausal women actually may have lost the reserve to maintain optimal progesterone levels and their feedback loop coordination is already compromised due to normal aging. The use of liquid progesterone may be very helpful in the management of their case. It is important to remember that in both menstruating women and menopausal women, the adrenal glands contribute to a significant portion of the total progesterone pools of the body and therefore the impact of adrenal function on progesterone status should always be investigated.

In adult women, severe hypothyroidism has been shown to cause elevations of prolactin and loss of ovulation. Therefore, insufficient progesterone may cause endometrial proliferation resulting in excessive and irregular breakthrough bleeding. The changes in progesterone receptor-site resistance and the loss of ovulation in severe hypothyroidism may reduce fertility and increase the risk of spontaneous abortion.

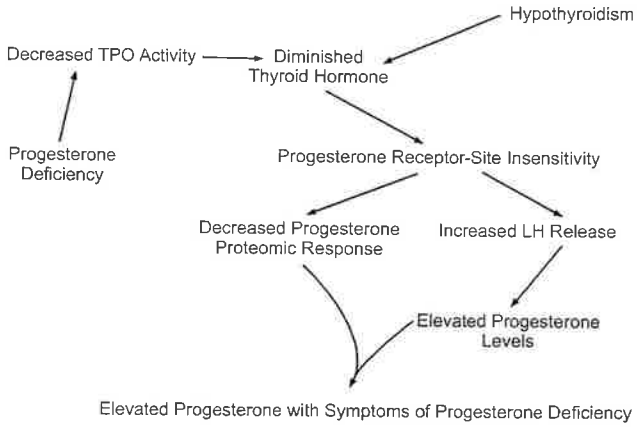
Not only does progesterone influence thyroid metabolism, but thyroid hormones impact progesterone receptor-site expression. Thyroid hormones such as thyroxine and triiodothyronine influence the sensitivity of progesterone receptor sites. When progesterone receptor sites are not exposed to an adequate amount of thyroid hormones, they lose their sensitivity to progesterone exposure. Clinically, patients may present with abnormal fluctuations and surges of progesterone on an expanded female hormone profile when they have patterns of hypothyroidism. It is always important to rule out thyroid dysfunction with abnormal patterns of progesterone.

Progestive Deficiency
For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- Progesterone levels

Progesterone-Thyroid Interplay



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Cortisol Metabolism and the Thyroid

In states of hypothyroidism, the urinary excretion of cortisol, 17-hydroxycorticosteroids and aldosterone is decreased. There have been no changes observed in the serum with these changes in hypothyroidism, so it appears to be a matter of clearance not synthesis. The use of urinary hormone tests for cortisol, 17-hyroxycorticostreroids, and aldosterone may not reflect levels found in the serum due to the influence of thyroid hormones on urinary excretion.[116]

Growth Hormone, IGF-1, and the Thyroid

Growth hormone (GH) is released by the anterior pituitary to promote anabolic expressions of metabolism. Growth hormone then stimulates the production of insulin-like growth hormone (IGF-1), also known as somatomedin in the liver. It is IGF-1 that is responsible for the many physiological impacts of GH on tissues. Although the human growth hormone gene does not contain a thyroid hormone response, the thyroid does influence the synthesis of IGF-1 in the liver. Adequate amounts of thyroid hormones are required for the healthy production of IGF-1. Anytime reduced IGF-1 markers are demonstrated, thyroid dysfunction must be ruled out. The reduced potential to synthesize IGF-1 may be partly due to the loss of anabolism in hypothyroidism.[117]

Summary of Endocrine-Thyroid Crosstalk in a Female

- Thyroid hormone synergizes with follicle-stimulating hormone (FSH) to exert stimulatory effects on granulosa cell differentiation and function, suggesting that it plays a physiological role in amplifying FSH-mediated differentiation of granulosa cells.
- Lack of thyroid hormones promotes the development of ovarian cysts.
- The estrous cycle become irregular in its distribution of estrogen, but it does not necessarily become deficient in hypothyroid states.
- Serum LH levels become reduced with hypothyroidism.
- In hypothyroid states, tissue progesterone receptor sites become insensitive to progesterone and at times show elevations on laboratory tests.
- Both FSH and LH releases become compromised in hypothyroid states.
- Menstrual cycle regulation with FSH and LH become altered in low thyroid states due to abnormal fluctuations of the pituitary hormones.
- Hypothyroidism significantly increases the pituitary content of vasoactive intestinal peptide (VIP) though it does not affect dopamine synthesis. These results suggest that hypothyroidism increases pituitary content of VIP, likely affecting prolactin secretion and leading to inhibition of gonadal function.
- Hypothyroidism leads to hypersecretion of endogenous CRH, leading to increased adrenal cortex activation.
- Even moderate hyperthyroidism produces an increase in glucose turnover and a concomitant acceleration in protein breakdown.
- Thyroid hormone is essential for glucose recycling.
- Adequate amounts of thyroid hormones are required for the healthy production of IGF-1.
- In states of hypothyroidism, the urinary excretion of cortisol, 17- hydroxycoricosteroids, and aldosterone is decreased.

Summary of Endocrine-Thyroid Crosstalk in a Male

- Low thyroid states result in depressed LH and FSH leading to potentially depressed testosterone and sperm synthesis from a pituitary-gonadal axis. Prolactin in not affected by low thyroid hormone status.
- Leydig cell enzyme activity of 3-beta and 17-beta hydroxysteroid dehydrogenase becomes inhibited, leading to decreased testosterone synthesis at the gonadal level.
- LH receptor sensitivity and LH receptor quantity in the gonads become insensitive in low thyroid status.
- Insensitivity develops with mRNS androgen receptor expression in low thyroid states.
- Hypothyroidism in males is associated with an increase in the circulating level of total cholesterol and LDL-cholesterol (LDL-C) and a reduction in the levels of progesterone and testosterone, without any change in the serum levels of oestradiol and gonadotropins.
- Thyroid hormone deficiency affects all tissues of the body, including multiple endocrine changes that alter growth hormone, corticotropin, glucocorticoids, and gonadal function.
- Hyperprolactinemia is noted in male hypothyroid patients and leads to an altered pituitary-gonadal axis.
- Hypothyroidism causes adrenal dysfunction directly and results in hypersecretion of corticotrophin-releasing hormone (CRH) and arginine vasopressin (AVP) from the hypothalamus, and the pituitary response to luteinizing hormone-releasing hormone (LHRH) is suppressed, whereas the testicular

response to hCG is not changed in hypothyroid subjects.

- Hypothyroidism leads to hypersecretion of endogenous CRH, leading to increased adrenal cortex activation.
- Even moderate hyperthyroidism produces an increase in glucose turnover and a concomitant acceleration in protein breakdown.
- Thyroid hormone is essential for glucose recycling.
- Adequate amounts of thyroid hormones are required for the healthy production of IGF-1.
- In states of hypothyroidism, the urinary excretion of cortisol, 17-hydroxycorticosteroids, and aldosterone is decreased.

Part II: Endocrine Activity on Thyroid Metabolism

In this section of the manual we review how non-thyroid hormones impact thyroid physiology. Hormones such as cortisol, estrogens, progesterone, and testosterone can have powerful impacts on thyroid physiology that are significant enough to cause thyroid-related disorders. These are considered NTI. In these mechanisms, the thyroid gland itself is not dysfunctional, but physiological alterations create a state of deficit thyroid hormone activity in the body, which in turn creates hypothyroid-related disorders. These mechanisms include altering thyroid-binding globulins, suppressing the pituitary-thyroid axis, altering T4 to T3 conversion, and creating thyroid receptor-site resistance.

The Impact of Prolactin on the HPT Axis

Prolactin is very sensitive to shifts of dopamine, thyroid hormones, progesterone, and serotonin. There is a very tight physiological loop between these hormones and neurotransmitters. A shift in one of these neurotransmitters or hormones can cause shifts in the others. Depressed dopamine and/or progesterone can promote elevations of prolactin that may suppress TSH functionally. On the other hand, a prolactinoma (benign pituitary tumor) may produce high amounts of prolactin in hypothyroid patterns. In either case, patients who have elevated prolactin may have a history of discharge from their breasts (both male and female). In females, elevated prolactin promotes suppression of luteinizing hormone (LH) and progesterone leading to infertility, dysmenorrhea, and amenorrhea. In men, elevated prolactin suppresses LH leading to depressed testosterone and symptoms of impotence and andropause.

If the elevated prolactin is caused by a tumor, it may promote severe headaches and visual disturbances (scotomas). Anytime serum prolactin is elevated, a prolactinoma must be ruled out. Imaging may be helpful, but many tumors cannot be visualized and not all prolactinomas are enlarged. Nutritional support should be focused on improving LH and dopamine but is not a substitute for medical attention.[67-69,118-119]

The Impact of Cortisol on the HPT Axis

Suppression of TSH may be caused by activation of the hypothalamus-pituitary-axis. The central integrative state of paraventricular (PVN) cells of the hypothalamus dictate the cortisol response. The cytokine messengers stimulate the PVN cells for cortisol responses, but they also exhibit suppressive influences on TSH. Therefore, it is not uncommon clinically to see patients with altered cortisol (hyperadrenal, hypoadrenal, and adrenal maladaptation) to present with suppressed TSH. The functional range of TSH is 1.8. Patients in this pattern will present with cortisol-related symptoms, a TSH below 1.8, a depressed functional T4 (<6), and thyroid symptoms. An adrenal salivary index demonstrates adrenal dysfunction. The main causes for this pattern are active infection, dysglycemia, or unrelenting stress. The most common cause is an uncontrolled blood sugar imbalance such as insulin resistance or hypoglycemia. If the patient feels shaky, irritable, and has sugar cravings between meals, suspect reactive hypoglycemia. If he or she has fatigue and sugar cravings after meals, suspect insulin resistance.[7-12]

Progesterone Influence on TPO Activity

The thyroid hormone-producing enzyme, thyroid peroxidase (TPO), is up-regulated with progesterone. That is why when women ovulate, their temperature becomes elevated. During ovulation, a surge of progesterone takes place that stimulates TPO. There is a surge of increased thyroid hormones produced that increases the metabolic rate and temperature. When patients suffer from progesterone insufficiency, they may develop less thyroid hormones than their bodies require, but not outside the laboratory ranges of TSH, T4, or T3. These patients typically complain of both low thyroid and low progesterone symptoms. Low progesterone symptoms and signs include heavy menstrual bleeding, migraines in the second half of the

menstrual cycle, infertility, depression, early miscarriage, breast tenderness, irregular periods, menstrual cramps, water retention before their period, and tendencies for ovarian cysts.

Most common causes for progesterone deficiency in a menstruating-age female is suppressed LH from stress and dysglycemia.

For the treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- Sugar metabolism
- The adrenals

The Impacts of Cortisol on 5'-Deiodinase Activity

Elevated cortisol has a suppressive impact on the enzyme 5'-deiodinase, which converts inactive T4 into active T3. Many sources have identified the pattern associated with elevated cortisol and its influence of low T3. The mechanism of lowered T3 may be due to multiple biochemical influences. Typically T4 is converted to T3, reverse T3, T3 sulfate (T3S), and triiodothyroacetic acid (T3AC) peripherally by the enzyme 5'-deiodinase. In the past, it was believed that the reason for the low T3 state was because it was shunted to the production of irreversible and inactive reverse T3. This explanation has recently proven to be incorrect since reverse T3 production rates measured in vivo are generally unaltered in stress patterns. Thus, it has been shown that reverse T3 concentrations result from diminished reverse T3 clearance rather than increased production. Current evidence published in *Thyroid response to critical illness* shows that increased levels of T3S and T3AC at the expense of T3 are most likely responsible for the lowered T3 state. T3S and T3AC are inactive forms of T3, but they have the potential to become activated into T3 by sulfation and acetylation. The bottom line is that elevated cortisol will decrease active T3 levels and therefore reduce the potential of optimal gene expression by thyroid hormones.[121-123]

The Impacts of Estrogens on Thyroid-Binding Globulin Levels

When thyroid-binding globulin levels increase, the percentage of free-fraction thyroid hormones drop. This may be observed by a depressed T3 uptake or a depression of free thyroid hormones and a normal TSH. This elevation is typically due to a source of exogenous estrogens. The most common cause of this pattern is from oral contraceptives and Premarin®.[124-128]

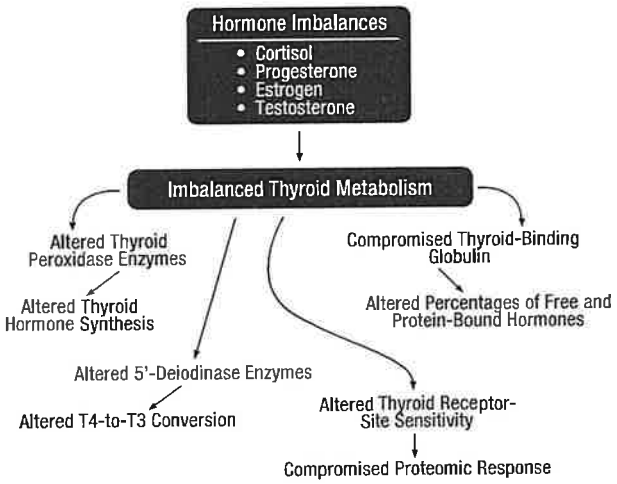
Please see registered trademark ownership in Appendix 1.

The Impacts of Testosterone on Thyroid-Binding Globulin Levels

When thyroid-binding globulin levels drop, the percentage of free thyroid hormones elevates. This may be observed by an elevated T3 uptake or elevated free thyroid hormones with a normal TSH. This elevation is typically subtle and does not create symptoms of hyperthyroidism. Instead, the individual develops symptoms of thyroid hypofunction due to thyroid receptor-site resistance. This pattern is typically caused by elevations of testosterone. Endogenous elevation in females is typically associated with metabolic syndrome. The constant unrelenting surges of insulin in metabolic syndrome stimulate the activity of 17,20-lyase in the thecal cells of the ovaries, which leads to elevations of testosterone. This pattern may also be promoted by exogenous overload of testosterone.[125-131]

The Impacts of Cortisol on Thyroid Receptors

Once thyroid hormones bind to receptor sites, a biochemical reaction called intercellular transduction is initiated. This intercellular transduction response carries the message of binding to nuclear receptors. Once the nuclear receptor has been activated, it will respond by producing proteins that express enhanced metabolic rate and energy production. Elevations in cortisol have been shown to decrease the expression of both alpha-1 and alpha-2 thyroid receptor-site responses.[132-134]



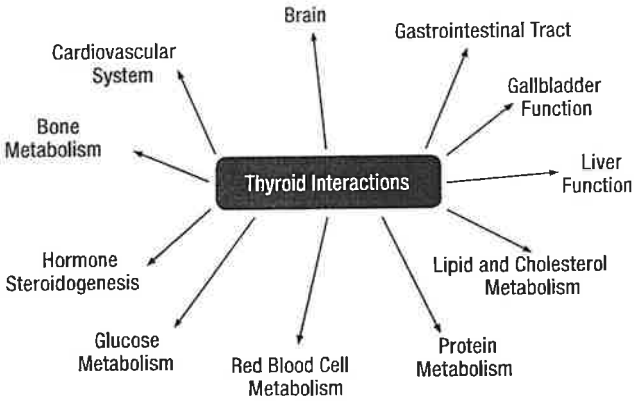
Part III: Thyroid-Endocrine-Thyroid Intertalk

Intimate relationships between the thyroid gland and male/female reproductive endocrine functions exist. Thyroid imbalances in men can lead to severe male hormone imbalances that include infertility, decreased testosterone synthesis, altered pituitary-gonadal axis function, etc. These changes may lead to conditions such as depression, erectile dysfunction, mood disorders, obesity, exhaustion, cognitive decline, etc. These clinical presentations are typically ignored because many of these symptoms will present over their thyroid symptomatology and typically the thyroid disorder is ignored. On the other hand, many andropause men suffer from cortisol and estrogen imbalances that may alter thyroid metabolism. Many men start taking exogenous testosterone that can also disrupt thyroid metabolism. An evaluation of these interrelationships should be considered with male hormone patterns.

In females, thyroid imbalances lead to severe defects in the pituitary-gonadal dysfunction that include alteration of FSH and LH coordination, suppressed release of FSH and LH, diminished ovarian FSH receptor-site sensitivity, and peripheral progesterone receptor sensitivity. These mechanisms lead to infertility, ovarian cysts, miscarriages, migraines, weight gain, depression, mood disorders, abnormal menstrual cycles, etc. Many times, the symptoms associated with these conditions outweigh the symptoms associated with thyroid symptomatology and the thyroid condition becomes overlooked. On the other hand, the use of exogenous hormones, such as birth control pills or hormone replacement therapy, in addition to insulin surges and stress responses, promote abnormal spikes of the hormones estrogen, testosterone, and cortisol, which disrupts healthy thyroid metabolism.

These endocrine-thyroid-endocrine relationships are common findings in clinical practice. Healthcare professionals must become familiar with these interconnections to be able to provide the proper advice and support to help restore ideal function to these individuals. Since the thyroid has such diverse impacts on endocrine physiology, and since thyroid metabolism is so sensitive to endocrine imbalances, a routing thyroid panel should be considered with all endocrine-related cases with or without the thyroid presentation symptomatology.

Thyroid Interactions



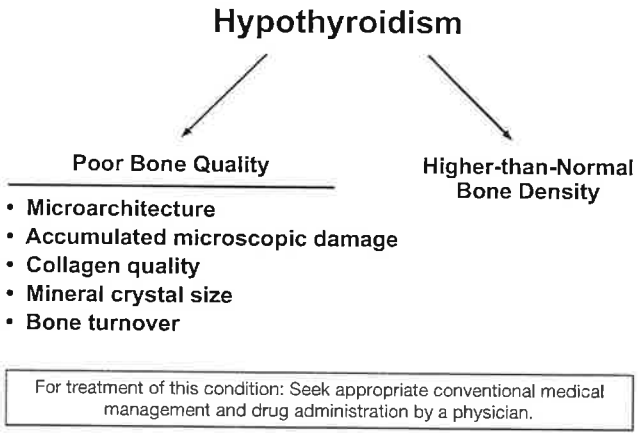
Thyroid Hormones and Bone Metabolism

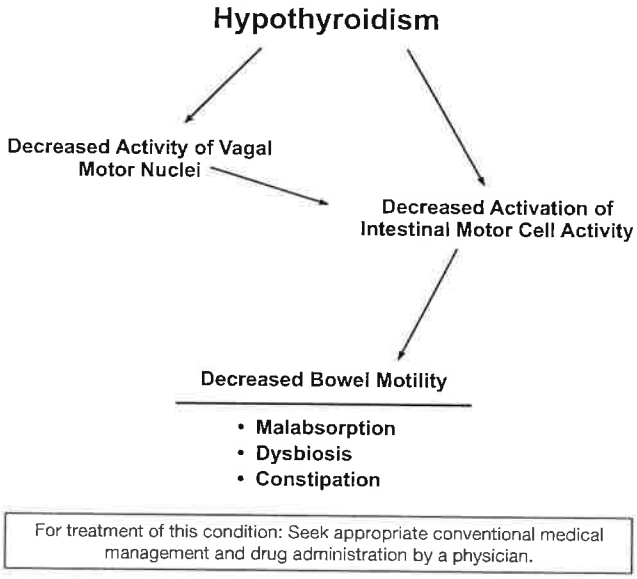
When evaluating the research on thyroid function and bone density, there are two separate issues to understand: the impact of thyroid states on bone density and the impacts of thyroid hormones on bone quality. "Studies of epidemiology of osteoporosis and of drug treatments for it have challenged the concept that denser bone means stronger bone. Bone strength or resistance to fracture is not easily measured by routine densitometry." Bone density and bone quality are not the same thing. "Most clinicians were taught directly or indirectly that bone density is the gauge for assessing bone strength and the response to anti-osteoporotic treatment. In recent years, however, the concept of bone strength has moved beyond density alone and has expanded to include a number of characteristics of bone that collectively are called quality." "**Bone quality** is a composite of properties that make bone resist fractures, such as microarchitecture, accumulated microscopic damage, the quality of collagen, mineral crystal size, and bone turnover." "Bone strength or resistance to fracture is more complex than density alone" because "denser bone is not always stronger," younger bone density is stronger than older bone density, and changes in density account for only part of the decrease in risk.[135]

"The mechanism of action as well as the clinical effects of thyroid hormones on bone has been of interest for more than a century. With the appearance of new treatment modalities for thyroid function disorders, the accompanying alterations in bone metabolism appeared to be rare. However, the introduction of new diagnostic tools, such as osteodensitometry and biochemical bone markers, has changed our view since they were able to show smaller but important changes in bone mineral metabolism associated with thyroid function disorders. Now, we know that hyperthyroidism is associated with an increased lifetime risk for fractures, even after achieving euthyroidism. This fact may play an important role in the higher mortality rate among previously hyperthyroid patients later in life. Subclinical hyperthyroidism may also affect bone density, however, its effect on fracture rate remains to be established. The lack of thyroid hormones will alter normal growth during childhood. Adult hypothyroid patients tend to exhibit higher than normal bone density. Despite increased bone quantity, hypothyroidism is accompanied by increased fracture risk before and after diagnosis. The effect of thyroid hormone treatment on bone tissue is somewhat controversial. Nevertheless, only patients with suppressed thyroid-stimulating hormone (TSH) appear to lose bone and have higher fracture incidence, whereas patients with a TSH level in the normal range seem to have similar bone mineral content and fracture rate as euthyroid subjects. In summary, most of thyroid function disorders may result in reduced bone density and/or increased fracture rate that should be taken into consideration at clinical evaluation."[136-137]

Summary of Thyroid–Bone Interactions

- Hyperthyroidism is associated with an increased lifetime risk for fractures, even after achieving euthyroidism.
- Lack of thyroid hormones will alter normal growth during childhood.
- Adult hypothyroid patients tend to exhibit higher-than-normal bone density. Despite increased bone quantity, hypothyroidism is accompanied by increased fracture risk before and after diagnosis.
- Patients with suppressed thyroid-stimulating hormone (TSH) appear to lose bone and have higher fracture incidences.
- Patients with a TSH level in the normal range seem to have similar bone mineral contents and fracture rates as euthyroid subjects.



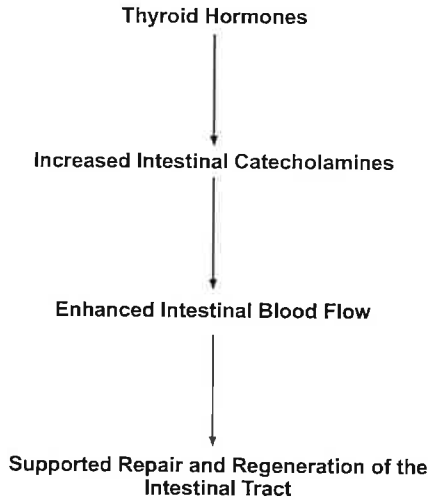


The Influence of Thyroid Hormones on Gastrointestinal Function

Thyroid diseases may be related to gastrointestinal motility symptoms. Such symptoms can vary in degree and, sometimes, are the only clue of a thyroid disease or, at least, the first. Hyperthyroidism and hypothyroidism can impair esophageal motility, modifying pharyngo-esophageal structure and/or muscular function and interacting with the neuro-humoral regulation of the esophageal peristalsis.

Both thyroid alpha-1 and beta-2 receptors are found in the dorsal motor nucleus of the vagus, nucleus tractus solitari, and area postrema. Thyroid hormone receptors indicate that thyroid hormone influences neuronal activity in these neuronal centers and contribute to visceral disorders.[138]

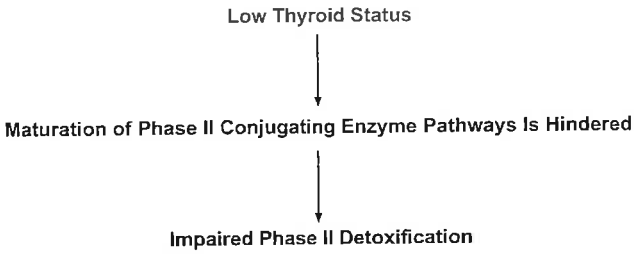
"At the gastroduodenal level, basic and postprandial electric rhythm alterations have been observed in hyperthyroid patients, often associated with delayed gastric emptying. In such patients, the autonomous nervous system dysfunction may even modify the neuro-hormonal mutual regulation (vagal influence decrease) of the gastroduodenal myoelectric activity. Hypothyroidism may cause a delay of gastric emptying. Diarrhea and malabsorption are common findings together with hyperthyroidism, whereas constipation is frequently observed in hypothyroidism."[139] The gastrointestinal transit time is often reduced in low-thyroid states. In addition, intestinal absorption rates are altered. The rates of absorption for substances are decreased but may be increased due to decreased bowel motility. Malabsorption and dysbiosis potential may exist in states of low thyroid function.[140]



Hepatic Function and Thyroid Hormones

Thyroxine (T4) secreted by the thyroid gland under normal physiological conditions is nondeiodinated in the liver to form 3,3'5-triiodothyronine (T3) before exerting biological activity at target tissues and receptors. Patients who suffer from hepatic cirrhosis have difficulty converting T4 into T3. Additionally, increased serum TSH is found in hepatic cirrhosis and is often accompanied by an abnormal TSH response to thyrotropin-releasing hormone suggesting, in addition, disordered hypothalamic-pituitary control of thyroid

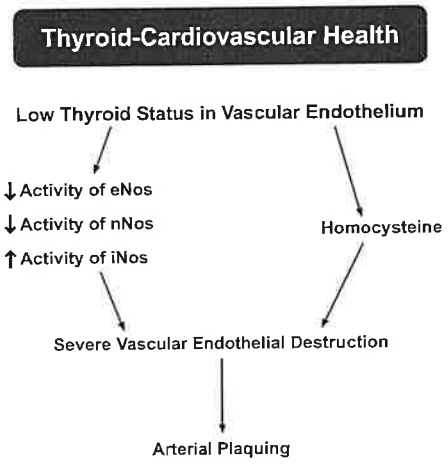
function.[141-142] Therefore, liver cirrhosis alters thyroid physiology at the central axis and at the 5'-deiodinase enzyme. It is likely that both of these effects are due to elevated cytokines as a consequence of cirrhosis. It has also been found that there is "diminished TSH response to TRH stimulation in patients with hepatic cirrhosis, despite subnormal serum T3 levels."[143]



Cardiovascular Risk and Hypothyroidism

"Thyroid hormone regulates several cardiovascular functions, and low T3 levels are frequently associated with cardiovascular diseases." A study found that "T3 exerts direct and acute effects on the resistance vessels by enhancing endothelial function and norepinephrine-induced vasoconstriction. The data may help clarify the vascular impact of low T3 syndrome and point to potential therapeutic strategies."[144]

"Subclinical hypothyroidism is associated with enhanced cardiovascular risk." Research has found that "patients with subclinical hypothyroidism are characterized by endothelial dysfunction resulting from a reduction in nitric oxide availability, an alteration independent of dyslipidemia and reversed by levothyroxine supplementation."[145]

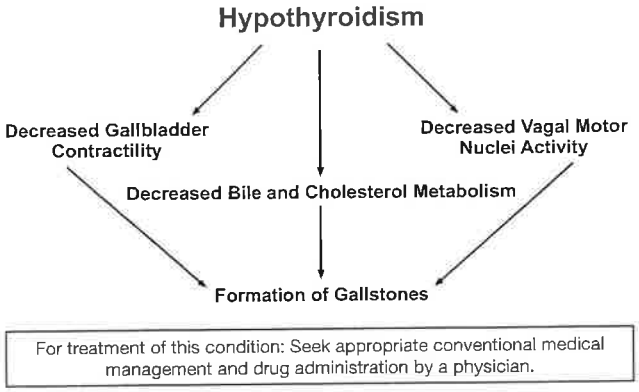


The Influence of Thyroid Hormones on the Gallbladder and Liver

In hypothyroidism, the liver enzymes are generally normal, but reduced hepatic clearance has been observed with elevations in the amino transaminases (SGOT, SGPT). Gallbladder function has also shown to be impaired in hypothyroidism—cholecystography often reveals a distended gallbladder that contracts sluggishly.[146]

Hypothyroidism and hyperthyroidism are known to affect cholesterol metabolism, bile acid synthesis, and bile content. Studies report increased prevalence of gallstone and common bile duct stone formation in hypothyroidism. Research has also shown an increased prevalence of previously diagnosed hypothyroidism in common bile duct stone patients and a delayed emptying of the biliary tract in hypothyroidism, explained partly by the missing pro-relaxing effect of thyroxine on the sphincter of Oddi contractility.[147] Disturbances in the sphincter of Oddi function have been shown to prevent normal bile flow and thus enhance the probability of common bile duct stone formation. Hypothyroidism has been shown to result in altered bile composition, delayed excretion, and emptying of the biliary tract.[148]

A study was conducted to investigate the frequency of diagnosed hypothyroidism in patients with common bile duct stones. The study found that there was a significant association between common bile duct stones and previously diagnosed hypothyroidism, and suggested all patients with common bile duct stones be screened for current thyroid dysfunction.[149]



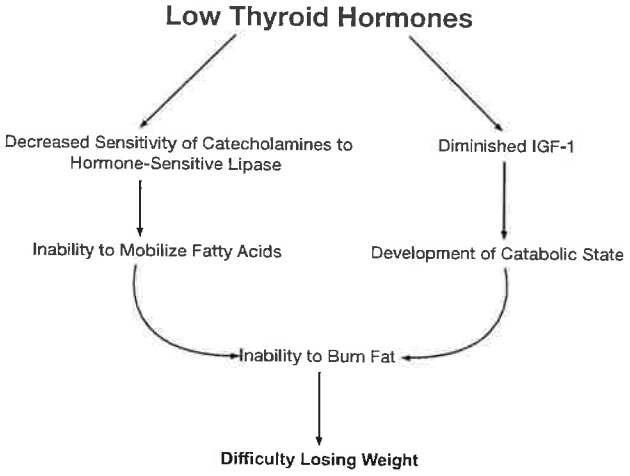
Hypothyroidism
For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- Liver enzymes and bile ducts

The Influence of Thyroid Hormones on Body Composition

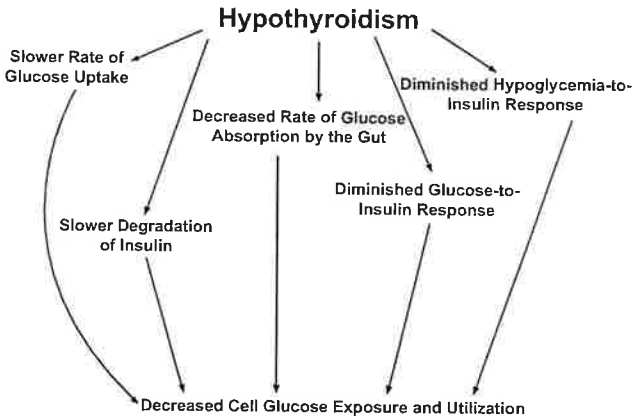
A common symptom of low thyroid status is an inability to lose weight. This is due to several factors. First, since thyroid hormones are responsible for metabolic activity, the loss of metabolic activity potential directly leads to an inability to lose weight and even to weight gain. Low thyroid status also decreases the mobilizations of free fatty acids, thereby hindering the response of catecholamines to lipolysis. Both of the above metabolic shifts also contribute to fatigue. A third pattern impacts body compensation related to growth hormone. In states of low thyroid status, if growth hormone response is hindered, there is greater potential to become catabolic and therefore lose muscle mass or lose the ability to gain muscle mass.[117]



The Influence of Thyroid Hormones on Insulin and Glucose Metabolism

In low thyroid status, the response of insulin to glucose is delayed. With oral glucose tolerance testing, there is a characteristic flat curve response to glucose stimulation in hypothyroidism because there is a slow rate of glucose uptake by tissues and a decreased rate of glucose absorption by the gut. In addition, in states of hypothyroidism, there is slower degradation of insulin; therefore, there is increased sensitivity to exogenous insulin. This response accounts for decreased exogenous insulin requirements when there is a presentation of coexisting hypothyroidism and insulin-dependent diabetes mellitus (IDDM).

The hindered response to glucose, the slow rate of glucose uptake, the decreased rate of glucose absorption, and the slower rate of glucose clearance present clinically as hypoglycemia. In hypothyroidism, despite glucose consumption, the ability for glucose to get into the cells is hindered; therefore, although the patient presents with symptoms of hypoglycemia, since the glucose levels are normal in the serum due to decreased uptake of cells, when a fasting glucose is conducted, they will not present as a hypoglycemic. This pattern leads to increased stress in the hypothalamus-pituitary-adrenal (HPA) axis because the adrenals will make attempts to release cortisol to increase the glucose load for cells that require it but are unable to utilize glucose efficiently. The result of hypothyroidism on glucose metabolism is decreased cell exposure with symptoms of hypoglycemia and up-regulation of the HPA axis. In addition, in states of low thyroid hormones, it has been found that the response of plasma cortisol to insulin-induced hypoglycemia may be impaired.[150]



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Thyroid Hormones, Lipid Metabolism, and Cholesterol

In hypothyroidism, both the synthesis and degradation of lipids are depressed, but the degradation of lipids is far slower than synthesis, so the net result is elevated total serum cholesterol, triglycerides, and LDL. The

decrease in lipid degradation rate may reflect the decrease in postheparin lipolytic activity, as well as reduced LDL receptors. In states of hypothyroidism, plasma free fatty acid levels are decreased. The mobilization of free fatty acids, in response to fasting, as well as catecholamines and growth hormone expressions, are impaired.

Anytime a lipid profile presents with hypercholesterolemia, the thyroid must be ruled out since it may be the culprit for the abnormal thyroid level. If an accompanying hypothyroid state is present with an abnormal lipid panel, management of the profile should be directly focused on the thyroid imbalance. Correction of the thyroid disorder will usually correct the abnormal lipid presentation.[151]

Hypothyroidism

↓

Diminished Degradation of Lipids

↓

Abnormal Lipid Panel

↑ Total Cholesterol

↑ LDL

↑ Triglycerides

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Catecholamines, Hypertension, and the Thyroid

"Arterial hypertension is known to be frequently associated with thyroid dysfunction, with a particularly high prevalence in chronic hypothyroidism. However, to our knowledge, no comprehensive study addresses causal mechanisms possibly involved in this association. We here report the physiological relationships between blood pressure and neurohumoral modifications induced by acute hypothyroidism in normotensive subjects. Twelve normotensive patients with previous total thyroidectomy were studied. Ambulatory 24-hour blood pressure monitoring was performed, and free T(3), free T(4), TSH, PRA, aldosterone, cortisol, adrenaline, and noradrenaline were assayed 6 weeks after oral L-T(4) withdrawal (phase 1) and 2 months after resumption of treatment (phase 2)." "In conclusion, our data show that thyroid hormones participate in the control of systemic arterial blood pressure homeostasis in normotensive subjects. The observed sympathetic and adrenal activation in hypothyroidism, which is reversible with thyroid hormone treatment, may also contribute to the development of arterial hypertension in human hypothyroidism."[152]

The Impact of Thyroid Hormones on Phase II Detoxification

Thyroid hormones have some degree of influence on all cells of the body, especially on hepatic cells responsible for enhancing the integrity of phase II detoxification. The liver is responsible for converting fat-soluble substances into water-soluble substances via phase I oxidation/reduction reactions and phase II conjugation reactions. Once compounds go through both phases, the compounds may be eliminated via the feces, sweat, or urine. Thyroid hormones are important in phase II metabolism because the maturation of phase II conjugating enzymes is influenced by thyroid hormones. In states of low thyroid hormone status, the maturation of phase II conjugating enzyme pathways may be hindered; therefore, detoxification may be hindered.[146]

Clinically, it is not uncommon to see patients who present with chronic thyroid disorders and have presentations of altered hepatic clearance potentials. When these patterns exist, nutritional support to improve hepatic clearance usually presents with poor clinical outcomes until the thyroid function is restored or replacement therapy is initiated.

The Influence of Thyroid Hormones on Hypochlorhydria

Gastrin is important for proper gastric function and the production of hydrochloric acid (HCl). In states of hypothyroidism, gastrin levels are usually reduced; therefore, concomitant hypochlorhydria is present. Hydrochloric acid is important for acidifying the bolus, which is then necessary to stimulate enteric reflexes involved in both pancreatic and bile output. Compromised thyroid function may have adverse reactions on healthy digestion. Anytime a pattern of hypochlorhydria is observed, thyroid dysfunction must be ruled out, and anytime low thyroid function is present, hypochlorhydria must be considered. Always use HCl support with these patterns until the thyroid problem is resolved.

A study was conducted "to investigate the effect of low circulating thyroid hormone levels on the development of acute stress gastritis in rats. Sixty adult Sprague-Dawley rats were divided into six groups: control group, surgically thyroidectomized group, stressed group, surgically thyroidectomized + stressed group, surgically thyroidectomized + T4 + stressed group, and surgically thyroidectomized + T3 + stressed group. Damage to the gastric mucosa was studied using millimetric acetate papers on photographs enlarged 3.5 times and the number and the size of the lesions was recorded." The results indicated that "acute stress gastritis was significantly increased in stress + surgically thyroidectomized rats as compared to rats that were only put under stress." The study concluded that "low circulating thyroid hormone levels in rats increased the development of stress gastritis. This effect could be prevented by thyroid hormone replacement therapy."[153]

The Influence of Thyroid Hormones on Protein Metabolism

In states of low thyroid function, the permeability to proteins is increased; therefore, an increase in the total concentrations of serum proteins may be increased. In addition, the albumin pools increase due to decreased albumin degradation as a consequence of thyroid hormones. The above influences of low thyroid status may create a false presentation of hypochlorhydria. In hypochlorhydria, a functionally elevated albumin and total protein may be present. The clinician must evaluate the patient's subjective indicators of hypochlorhydria to distinguish from a true HCl need or from secondary shifts due to a thyroid imbalance.

Note that in most cases, the patient does not have a true hypochlorhydria pattern that is positively supported with HCl supplementation.

Thermoregulation, Hot Flashes, and Thyroid Hormones

Since the thyroid modulates basal metabolic activity in patterns of thyroid insufficiency, patients may exhibit symptoms of hot flashes, night sweats, and abnormal thermoregulation. This becomes especially complex in women who have perimenopause in adjunct with a thyroid disorder. Many times, clinicians assume their symptoms are due to a loss of optimal estradiol output that accompanies menopause, however, the symptoms may in fact be due to an underlying thyroid disorder, or both. Hormone profiles are especially important in these cases. Remember, three main glands have an influence on thermoregulation: thyroid, ovarian, and adrenal. The symptoms of altered thermoregulation from ovarian influence versus thyroid influence may be hard to discern in perimenopause, however the clinician would expect to see other expressions of thyroid symptoms if the thyroid was involved. With adrenal influences of hot flash presentations, the patient usually does not notice major changes in temperature fluctuations, but rather is just experiencing sweating attacks confused as hot flashes. This is caused by mineralocorticoid shifts that take place with adrenal dysregulation.[154]

The Influence of Thyroid Hormones on the Hematopoietic System and Anemia

The hematopoietic system has the potential to become compromised in different ways with states of low thyroid activity. Red blood cell mass becomes decreased in response to decreased production of erythropoietin and may present a pattern of normocytic normochromic anemia.

Other times, a macrocytic anemia may present itself as a consequence of low thyroid activity. The macrocytic anemia may be related to decreased absorption of B12 and folic acid from the concomitant shifts of hypochlorhydria induced by hypothyroidism. Research has shown that achlorhydria after maximal histamine stimulation may be present with patients with primary hypothyroidism. A macrocytic anemia may also be present in association with pernicious anemia. Overt pernicious anemia is present and reported in about 12% of patients with hypothyroidism and 9% of patients with hyperthyroidism. The coexistence of pernicious anemia, which is an autoimmune response against intrinsic factor, and the presence of thyroid disorders with positive thyroid antibodies supports the view that the pathogenesis of the pattern as an overzealous immune disorder. To distinguish a B12 macrocytic anemia due to malabsorption or inadequate diet from an autoimmune pernicious anemia, the clinician must request an intrinsic factor autoantibody test (IF Ab). If the IF Ab is positive, then the diagnosis of pernicious anemia may be made.

Since hypothyroidism is associated with altered iron absorption secondary to hypochlorhydria and menorrhagia from progesterone receptor-site resistance, a possibility of a microcytic hypochromic anemia is also present.

The astute clinician should always screen for thyroid disorders with abnormal CBC patterns, and always screen for hematopoietic shifts with thyroid disorders. If someone presents with patterns of normocytic normochromic anemia, treatment should be focused on the thyroid. If he/she presents with macrocytic anemia, it is important to rule out the diagnosis of pernicious anemia by requesting an IF Ab test. With a nonpernicious anemia, B12 and folate supplementation—in conjunction with HCl support—should be considered. With pernicious anemia, the patient should be supported with liquid B12 and clinical management related to autoimmune disorders. If the patient presents with microcytic anemia, iron supplementation in conjunction with HCl should be considered until the thyroid is restored.

The Influence of Thyroid Hormones and Sex Hormone-Binding Globulin

Research has shown that sex hormone-binding globulin (SHBG) levels may be reduced in states of thyroid deficiency. SHBG levels have an influence on the free-fraction states of steroid hormones such as estradiol. Remember, hormones bind to SHBG to travel around in the tissue in an inactive state. When a hormone binds to its receptor site, it must lose its binding protein; therefore, thyroid hormones can influence the amount of protein binding and therefore the amount of free fraction hormones available. These shifts may influence test results of hormones strongly associated with SHBG—estradiol when compared in a free-fraction state, such as saliva, or in a protein-bound state, such as serum estradiol.

The Influence of Thyroid Hormones on Homocysteine and Methylation

When there is low thyroid hormone tissue exposure, it appears that there is altered methylation reactions, as evidenced by an increase in serum homocysteine levels. Homocysteine elevations have been shown to be an associated risk factor for cardiovascular disease, dementia, neurodegenerative diseases, cervical dysplasia, etc. Anytime homocysteine is elevated, use a high-level methyl donor.

Homocysteine and Methylation Dysregulation
For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- Homocysteine and methylation levels

Thyroid Influences on Epinephrine

Not only do neurotransmitters impact the HPT axis, but thyroid states also influence the production of neurotransmitters in the central nervous system. Central nervous system catecholamines in hyperthyroid, hypothyroid, and euthyroid states appear to have defined differences. Increased levels of epinephrine, 3,4-dihydroxymandelic acid (metabolite of brain norepinephrine), metanephrine, and homovanillic acid were found in hyperthyroid states compared to euthyroid subjects.

The levels of norepinephrine, dopamine, and normetanephrine were decreased in hyperthyroid states compared to euthyroid subjects. The levels of dihydroxymandelic acid, norepinephrine, and normetanephrine were all elevated in hypothyroid states compared to euthyroid subjects.[155]

The responsiveness of the cAMP pathway to epinephrine appears to be hindered with low thyroid hormone states; therefore, decreased adrenergic activity may be present. The mechanism underlying the adrenergic response is uncertain. A loss of adrenergic response may be the underlying mechanism in depression, weight gain, mood disorders, and reduced initiative.[156]

TRH Suppression with Thyroid Hormone Replacement

Thyroid hormone replacement therapy not only leads to TSH suppression, but also to suppression of TRH from the paraventricular nucleus of the hypothalamus, as evident by numerous studies.[157-160] This is important because TRH has independent impacts on physiology unrelated to its role in activating the pituitary-thyroid axis. Thyroid-releasing hormone also has peripheral autonomic influences through its role in regulating parasympathetic and sympathetic preganglionic motor neurons.[161] These findings may explain the clinical

presentation of altered autonomic activity in thyroid disorders, especially occurring after taking exogenous thyroid hormones leading to TRH suppression.

Exogenous thyroid hormone replacement leads to an independent side effect of suppressing TRH and altering the independent impacts TRH has on physiological systems. TRH is a powerful neuropeptide that has its own diverse set of actions on central structures of the brain stem necessary for autonomic and visceral function, which include activation of digestive enzymes, supporting gastrointestinal motility, regulating peripheral visceral nitric oxide and prostaglandin activity, promoting organ blood flow, supporting insulin release, modulating gallbladder contractility, etc. In this section of the manual we will review some of the most critical issues involved in independent TRH neuropeptide activity in the brain stem and the consequences of its lack of activation in peripheral viscera.

TRH and Brainstem Autonomic Activity

TRH and Brainstem Autonomic Activity – Gastric

"Thyrotropin-releasing factor may thus act as a neurotransmitter or neuromodulator in the dorsal brainstem and may participate in the regulation of autonomic functions.[162] TRH causes direct excitation of dorsal vagal and solitary tract neurons in brainstem slices."

Research has demonstrated the "excitatory action of thyrotropin-releasing hormone on the dorsal motor nucleus of the vagus neurons, leading to stimulation of vagal outflow to the stomach, and to vagal cholinergic-dependent stimulation of gastric secretory and motor function and alterations of mucosal integrity in rats and cats." The evidence suggests that TRH plays a physiological role in mediating vagal activity in the cephalic phase of digestion.[163]

Specific neuropeptides are important chemical messengers involved in physiologic brain processes regulating gastric acid secretion; they are involved in the intimate brain-gut relationship. Brain alterations of catecholaminergic, serotonergic, dopaminergic, GABAergic, and cholinergic pathways have been shown to significantly affect gastric acid secretion.[164] Thyrotropin-releasing hormone has been shown to stimulate release of gastric nitric oxide through vagal nicotinic pathways and peripheral activation of muscarinic receptors through TRH-sensitive vagal pathways.[165] Prostaglandin E2 is used in the gastric mucosa to modulate intestinal regeneration, gastric acid release, and modulation of inflammatory responses. Thyrotropin-releasing hormone has been shown to play a role in central regulation of gastric PGE2 release.[166] "The vagus nerve plays an important role in gastric cytoprotection and gastric lesion formation, but the central and peripheral mechanisms by which the vagus mediates these dual effects on the gastric mucosa are still not well established." Thyrotropin-releasing hormone has been shown to "stimulate gastric vagal efferent discharges. Such increased vagal activity results in cholinergic stimulation of the gastric secretory and motor functions that influence the resistance of the gastric mucosa to injury." "This mucosal resistance is influenced by the interaction between vagal-cholinergic activation of cytoprotective mechanisms (prostaglandin, increased gastric mucosal blood flow) and ulcerogenic factors (histamine, acid, pepsin, motility modulation)."[167]

TRH and Brainstem Autonomic Activity – Intestinal

"Central vagal activation by TRH stimulates intestinal mast cell secretion, in part via peripheral muscarinic receptors, and is modulated by PGs and capsaicin-sensitive afferent innervation. Intestinal mast cell activation contributes to the TRH analogue-stimulated luminal protein release." These mechanisms support intestinal permeability and immune coordination in the intestines.[168]

"Neurons in the dorsal vagal complex, including the dorsal motor nucleus (DMN) of the vagus, express TRH receptor and are innervated by TRH fibers." "TRH injected into the DMN or cisterna magna increases the firing of DMN neurons and gastric vagal efferent discharge, activates cholinergic neurons in gastric submucosal and myenteric plexuses, and induces a vagal-dependent atropine-sensitive stimulation of gastric secretory (acid, pepsin) and motor functions" necessary for intestinal motility.[169]

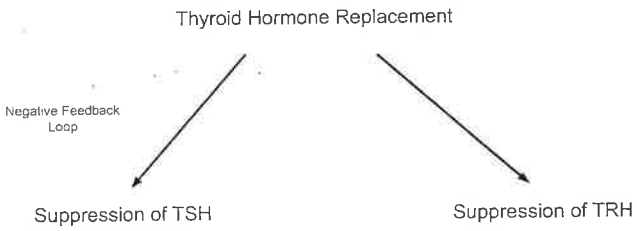
TRH and Brainstem Autonomic Activity – Pancreatic

"Pancreatic insulin secretion is regulated by the vagus nerve. Medullary thyrotropin-releasing hormone TRH containing projections from the raphe pallidus (Rpa) neurons innervate vagal preganglionic motor neurons in the dorsal vagal complex (DVC) and are involved in vagal regulation of gastric functions."[170]

TRH modulates calcitonin gene-related peptides on exocrine pancreatic secretion. The effects of TRH are mediated by vagal efferents, and the primary peripheral transmitter appears to be vasoactive intestinal peptide.[171] Research has found that "TRH acts in the central nervous system to stimulate pancreatic blood flow through vagal-cholinergic and nitric oxide-dependant pathways."[172]

Negative Feedback Loop

Thyroid Replacement with TSH and TRH



TRH and Brainstem Autonomic Activity – Hepatic

"Central neuropeptides play a role in many physiological regulatory processes through the autonomic nervous system. Thyrotropin-releasing hormone is distributed in the central nervous system and acts as a neurotransmitter to regulate hepatic functions through vagal-cholinergic pathways."

"Thyrotropin-releasing hormone acts in the central nervous system to stimulate hepatic blood flow through vagal-muscarinic and indomethacin- and nitric oxide-dependent pathways." Hepatic blood flow and prostaglandin are well recognized as cytoprotective factors for liver damage.[173] An animal study injected male rats with tetrachloride to induce hepatocellular necrosis and TRH injections were used after chemical destruction and compared to controls. Thyrotropin-releasing hormone was found to have cytoprotective effect on acute necrosis mediated through vagal-muscarinic, prostaglandin, and nitric oxide-dependent pathways.[174]

TRH is a central peptide that has been shown to modulate the autonomic nervous system and is involved in hepatic DNA synthesis cell regeneration. The results of a 1997 study "indicate that TRH acts in the CNS to stimulate hepatic DNA synthesis through vagal and cholinergic mechanisms, and that TRH may be the chemical messenger involved in brain regulation of hepatic proliferation."[175]

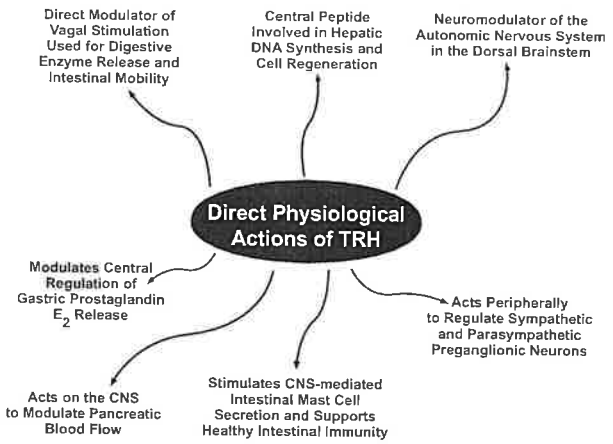
Central administration of TRH enhances hepatic blood flow. TRH acts on the left dorsal vagal complex to stimulate hepatic blood flow through the left cervical and hepatic vagus, indicating that neuropeptides may act in the specific brain nuclei to regulate hepatic function.[174]

TRH and Brainstem Autonomic Activity – Gallbladder

Cerebral TRH has been shown to modulate gallbladder contraction due to its influence on vagal-gallbladder related pathways and release of vasoactive intestinal peptide acting via its specific receptor. Cerebral calcitonin gene-related peptide pathways inhibit gallbladder contraction by stimulation of sympathetic, noradrenergic outflow via alpha-adrenergic receptors. Therefore, TRH suppression or dysregulation may lead to dysregulated autonomic control of gallbladder function.[176] TRH central stimulation has been shown to increase blood flow to the liver by capsaicin-sensitive afferent neurons and calcitonin gene-related peptide.[177]

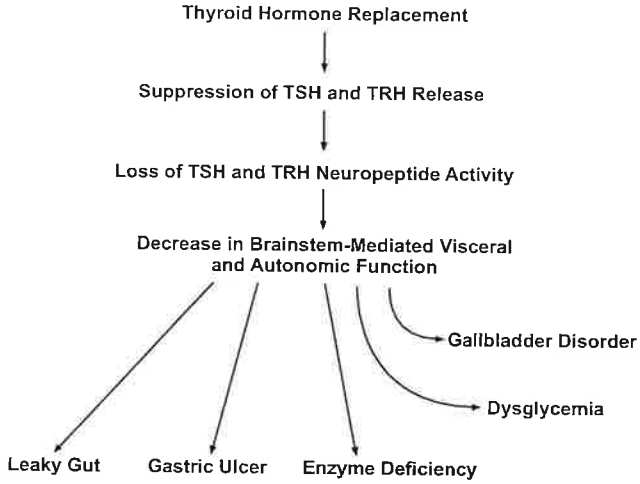
TSH and Brainstem Autonomic Activity – Vascular

A study was conducted to evaluate the impacts of TSH on vascular endothelial cells and vascular smooth muscles that possess TSH receptors. "Two different double blind, controlled studies were performed. Recombinant human TSH was infused intrabrachially to raise TSH to severe hypothyroidism levels, one in eight healthy volunteers and the other in six thyroidectomized patients. Endothelium-dependent and endothelium-independent vasodilation was tested by intra-arterial infusion of acetylcholine and sodium nitroprusside, respectively, and forearm blood flow was measured by plethysmography." The results found that "recombitant human TSH potentiated endothelium-dependent vasodilation and it was concluded that TSH exerts marked effects on the resistance vessels by enhancing endothelial-mediated vasodilatation, independent of changes in thyroid hormone concentrations."[17]

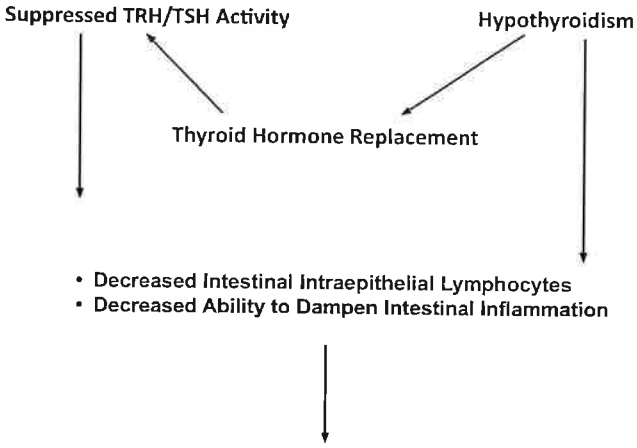


Summary of Direct Physiological Actions of TRH

- Is a neuromodulator of the autonomic nervous system in the dorsal brainstem
- Is a direct modulator of the dorsal motor nucleus in the brain stem and is involved with the vagal stimulation of cholinergic gastric secretory functions of producing digestive enzymes and motor functions of intestinal motility
- TRH acts peripherally to regulate parasympathetic and sympathetic preganglionic motor neurons
- TRH acts on the central nervous system to stimulate pancreatic blood flow through vagal-cholinergic and nitric oxide pathways
- TRH modulates brainstem activated pancreatic insulin modulation and release
- TRH has been shown to play a role in central regulation of gastric prostaglandin E2 release
- TRH acts as a central peptide and is involved with hepatic DNA synthesis and cell regeneration
- TRH acts on the vagal brainstem areas to stimulate intestinal mast cell secretion and support healthy intestinal permeability and immune coordination in the intestines
- TRH acts on the vagal brainstem areas to modulate gallbladder contractions due to its influence on vasoactive intestinal peptide
- TSH has been shown to promote vascular endothelial cells smooth muscle vasodilation independent of influences from thyroid hormones

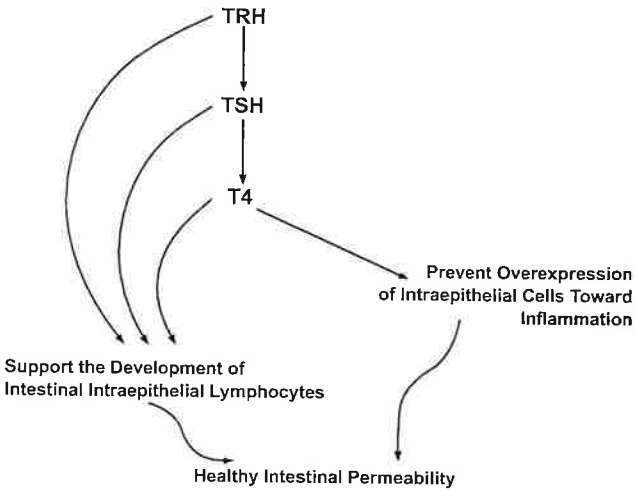


For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.



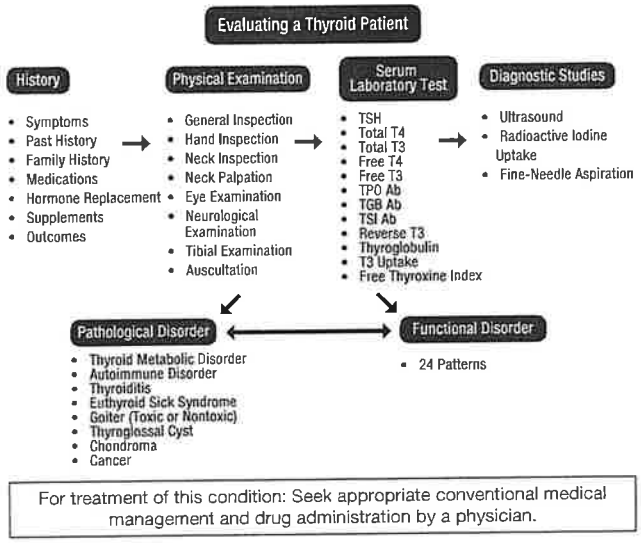
Leaky Gut Syndrome

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.



Thyroid Examination

A comprehensive thyroid physical examination is necessary for two reasons. First, the examination findings will help determine if an individual is suffering from a potential thyroid disorder. Second, the examination findings will help determine which systems the thyroid disorder has impacted. Common areas of impact include the skin, nails, fingers, hair, eyes, body composition, voice, fingers, throat, mood, brain, peripheral nerves, autonomic nervous system, cardiovascular system, and the gastrointestinal system. The ability to identify physical examination findings is sometimes the only clinical finding used to identify a thyroid disorder in asymptomatic cases.



Thyroid Examination Findings

Myxedema: a skin and tissue disorder usually due to severe prolonged hypothyroidism. It is characterized as jelly-like infiltrations in subcutaneous tissues, skin thickening, coarse skin, dry skin, yellow skin, non-pitting edema, and eye puffiness. It is typically found on the tibia and/or face, which are referred to respectively as *pretibial myxedema* or *facial myxedema*. It may occur anywhere on the skin. Myxedema may stem from both hyper- and hypothyroid conditions. It results from the accumulation of increased amounts of hyaluronic acid in the chondroitin sulfate in the dermis in both lesional and normal skin. The mechanisms that cause myxedema are still not understood, although animal model studies suggest that thyroid hormones affect the synthesis and catabolism of mucopolysaccharides and collagen by dermal fibroblasts. Treatment of myxedema is difficult even after normal thyroid function has been achieved.

Peaches-and-Cream Complexion: characterized by patches of thickened yellowish skin and discoloration and nonconsistent skin pore sizes. This is caused by the influence of thyroid hormones on collagen metabolism and typically found with hypothyroidism.

Acropachy: characterized by soft tissue swelling and periosteal bone formations, which usually causes clubbing in the fingers, toes, and lower extremities. It is usually the consequence of increased thyroid hormone responses in cartilage metabolism. Thyroid acropachy is usually seen in patients with associated pretibial myxedema and thyroid eye muscle disorders.

Onycholysis (Plummer's nails): a "loosening of the exposed portion of the nail from the nail bed, usually beginning at the free edge and continuing to the lunula." This is caused by the influence of increased thyroid hormones on collagen metabolism.

Palmar and Plantar Erythema: "a reddening of the skin on the palmar aspect of the hands, usually over the hypothenar eminence. It may also involve the thenar eminence and fingers. It can also be found on the soles of the feet when it is termed planter erythema." It may be seen with hyperthyroidism as well as many other diseases. It may be caused by "several underlying mechanisms such as increased cardiac output, capillary dilation in the palms due to local factors, localized inflammation, etc." In addition to thyroid hyperfunction, it may be found in liver diseases, pregnancy, connective tissue disorders, reflex sympathetic dystrophy, etc. (Please refer to Patient.co.uk [www.patient.co.uk/doctor/palmar-erythema] for more information.)

Thyroglossal Cyst: "a fibrous cyst that forms from persistent thyroglossal duct" during embryological development for a small portion of the population and presents as a midline neck lump. "Thyroglossal duct cysts most often present with a palpable asymptomatic midline neck mass below the level of the hyoid bone. The mass on the neck moves during swallowing. Diagnostically, a thyroglossal cyst will move upward with protrusion of the tongue." Some patients will have neck or throat pain, or dysphagia. "The persistent duct or sinus can promote oral secretion, which may cause

cysts to become infected. Up to half of thyroglossal cysts are not diagnosed until adult life. The tract can lie dormant for years or even decades until some kind of stimulus leads to cystic dilation. Infection can sometimes cause the transient appearance of a mass or enlargement of the cyst, at times with periodic recurrences. Spontaneous drainage may also occur."

Treatment for a thyroglossal cyst is surgical resection, often requiring concomitant removal of the midsection of the hyoid bone. "Although generally benign, the cyst will be removed if the patient exhibits difficulty in breathing or swallowing, or if the cyst is infected. Even if the symptoms are not present, the cyst may be removed to eliminate the chance of infection or development of a carcinoma, or for cosmetic reasons if there is an unsightly protrusion from the neck."

Proptosis: "Some sources define exophthalmos as a protrusion of the globe greater than 18 mm and proptosis as a protrusion equal to or less than 18 mm. Others define 'exophthalmos' as protrusion secondary to endocrine dysfunction and 'proptosis' as any nonendocrine-mediated protrusion." It is important for the clinician to not always jump to the diagnosis of endocrine disorders when seeing anterior protruding eyes. A proper history and examination can help define proptosis from exophthalmos. (Please refer to AskDefine [exophthalmos.askdefinebeta.com] for more information.)

Exophthalmos: "a bulging of the eye anteriorly out of the orbit. Exophthalmos can be either bilateral (as is often seen in Graves' disease) or unilateral (as is often seen with orbital tumor)." In thyroid disease, excessive thyroid hormones cause hypertrophy of the ocular muscles as well as promote excessive connective tissue deposition in the orbits. The tissue changes of exophthalmos are permanent and will not restore to normal even after thyroid dysfunction has been managed. "If left untreated, exophthalmos can cause the eyelids to fail to close during sleep, leading to corneal dryness and damage. The process that is causing the displacement of the eye may also compress the optic nerve or ophthalmic artery leading to blindness."

Diplopia: "commonly known as double vision or a simultaneous perception of two images of a single object." This may be present in hyperthyroid diseases due to the influence of thyroid hormones on ocular muscle atrophy and connective tissue deposits in the orbit.

Lid Lag (Graefe's sign): a condition where eyelid movement is delayed as the eye moves downward. This is identified by having the patient track a finger downward while observing for synchronous lid and eye movements. The movement of the eyelid downward should be in synchrony with the movement of the pupil. Lid lag is caused by excessive contraction of the levator palpebrae muscle from the influence of increased thyroid hormones on muscle metabolism.

Dalrymple's Sign: "widened palpebral (eyelid) opening or eyelid spasm, seen in thyrotoxicosis (as seen in Graves' disease but also other hyperthyroid conditions), causing abnormal wideness of the palpebrae fissure. With retraction of the upper eyelid, the white of the sclera is visible at the upper margin of the cornea in direct outward stare."

Stellwag's Sign: rare blinking caused by excessive contraction of the levator palpebral muscle from the influence of increased thyroid hormones on muscle metabolism. Patients appear as if they are always staring because they are not blinking.

Rosenbach's Sign: tremor of the eyelids caused by excessive contraction of the levator palpebral muscle from the influence of increased thyroid hormones on muscle metabolism. This is best evaluated by having patients close their eyes. Increased mesolimbic activity of the brain found in certain dystonic neurological disorders may also cause tremor of the eyelids, referred to as blepharospasm.

Jellinek's Sign: is hyperpigmentation of the eyelids due to the influence of thyroid hormones on increased production of melanin.

Examination Findings of Hyperthyroidism and Hypothyroidism

General Appearances

- General Appearances – Hyperthyroidism
  - Swollen fingers
  - Clubbing of fingernails
  - Purple lesions on shins
  - Proptosis or exophthalmos
  - Goiter
  - Normal body weight with potential muscle wasting

- General Appearances – Hypothyroidism
  - Peaches-and-cream complexion
  - Swelling of facial tissue, puffy eyes
  - Deep voice
  - Goiter
  - Obesity (normal body weight with Hashimoto's hypothyroidism)

Psychological Findings

- Psychological Findings – Hyperthyroidism
  - Irritability
  - Behavioral changes
  - Restlessness
  - Malaise
  - Anxiety
  - Excessive appetite
- Psychological Findings – Hypothyroidism
  - Depression
  - Poor mental endurance and brain fatigue
  - Poor memory
  - Loss of alertness
  - In extreme cases, psychosis and dementia
  - Loss of appetite

Dermatological Findings

- Dermatological Findings – Hyperthyroidism
  - Onycholysis
  - Palmar or plantar erythema
  - Facial or pretibial myxedema
  - Sweaty palms
  - Oily skin
  - Hyperpigmentation of the eyelids (Jellinek's sign)
- Dermatological Findings – Hypothyroidism
  - Dry, brittle hair
  - Alopecia
  - Dry, coarse skin
  - Xeroderma (dry skin)

Musculoskeletal Findings

- Musculoskeletal Findings – Hyperthyroidism
  - Proximal muscle wasting of shoulders and hips
  - Muscle weakness and poor muscle endurance
  - Hyperkinesis (overactive and restless body movements)
  - Tremor of hands and levator palpebrae muscles
  - Proptosis or exophthalmos
- Musculoskeletal Findings – Hypothyroidism
  - Muscle weakness and poor muscle endurance
  - General lethargy
  - Carpal tunnel syndrome

Neurological Findings

- Neurological Findings – Hyperthyroidism
  - Excessive activity of levator palpebrae muscles
    - Graefe's sign (lid lag)
    - Dalrymple's sign (widened eyelid opening)
    - Stellwag's sign (rare blinking)
    - Rosenbach's sign (tremors of eyelids)
  - Tremors of hands
  - Overactive deep tendon reflexes
  - Diplopia (due to hypertrophy of ocular muscles)
- Neurological Findings – Hypothyroidism
  - Ptosis (drooping of the eyelids)
  - Slow deep tendon reflexes
  - Ataxia and coordination difficulty

Cardiovascular/Autonomic Findings

- Cardiovascular/Autonomic Findings – Hyperthyroidism
  - Heat intolerance
  - Atrial fibrillation
  - Tachycardia
  - Cardiac palpitations
  - Cardiac failure
  - Systolic hypertension
  - Warm vasodilator peripheries
  - Respiratory palpitations/breathlessness
- Cardiovascular/Autonomic Findings – Hypothyroidism
  - Cold intolerance
  - Bradycardia
  - Hypertension/hypotension
  - Heart failure
  - Pericardial effusion

Gastrointestinal Findings

- Gastrointestinal Findings – Hyperthyroidism
  - Diarrhea
  - Vomiting
  - Excessive bowel frequency

- Gastrointestinal Findings – Hypothyroidism
  - Constipation
  - Floating stool
  - *Candida* syndromes
  - Malabsorption and malnutrition

The Steps of Conducting a Thyroid Physical Examination

General Inspection

- How are patients dressed to compensate for abnormal thermoregulation?
- Are they anxious or agitated?
- Are they overweight, normal weight, or underweight?
- How is their muscle mass?
- Do they have facial myxedema?
- Do they have a peaches-and-cream complexion?
- Do they have hair thinning or loss on scalp?
- Do they have loss of hair in outer third of eyebrow?

Hand Inspection

- Is thyroid acropachy (clubbing) present?
- Is there onycholysis?
- Is there palmar erythema?
- Are the palms sweaty or dry?
- Does the pulse demonstrate tachycardia, bradycardia, or atrial fibrillation?

Neck Inspection

- Are there any goiters or midline masses present?
- Have the patient swallow during inspection to observe masses.
- If mass in midline is noted, have the patient protrude his or her tongue. Movement of the midline mass with tongue protrusion indicates thyroglossal cyst.

Neck Palpation

- Palpate the thyroid gland for masses passively and while the patient is swallowing.
- Palpate for salivary gland lymphadenopathy.
- Auscultate the thyroid for bruit on both sides. Have the patient take a breath and hold it during auscultation. Hyperthyroidism may result in hyperdynamic circulation, tachycardia, and hypertension. The bruit may not indicate carotid artery stenosis. A carotid artery ultrasound should be considered if a bruit is present.
- If the thyroid gland is enlarged, percuss the retrosternal extension. Goiters may invade the chest cavity leading to abnormal retrosternal percussion. A flat or dull sound may indicate a goiter that has infiltrated the chest cavity.

Eye Examination

- Perform cardinal fields of view to assess for diplopia.
- Perform vertical eye pursuits to look for lid lag.
- Look at the patient from above and the sides to observe for proptosis and exophthalmos.

Motor Examination

- Assess deep tendon reflexes.
- Assess proximal muscle weakness.

Tibial Examination

- Examine for pretibial myxedema.

Auscultation

Bruit (bru-we): "is an abnormal and often harsh sound heard with a stethoscope over the blood vessel, usually an artery. The sound, which is synchronous with the heartbeat, is caused by turbulent blood flow, such as caused by irregularities in the wall of the blood vessel or by hyperdynamic circulation."

1. Thyroid Gland: Listen for thyroid bruit. A thyroid bruit is present at times in hyperactive thyroid conditions. The bruit is cause by increased blood flow to the thyroid.
2. Carotid Artery: Listen for carotid artery bruit. A bruit may be present with hyperactive thyroid disorders from hyperdynamic circulation. Other causes of carotid artery bruit include atherosclerotic plaque or stenosis. Carotid bruit should be followed by a carotid ultrasound.
3. Heart: Listen for tachycardia (hyperthyroid), bradycardia (hypothyroid), murmurs, splitting, irregular rhythm associated with influences of thyroid hormones on cardiac output and autonomic regulation. Abnormal heart sounds should be followed up by a cardiologist and an EKG.
4. Abdomen: Listen for bowel sounds. Normal sounds consist of clicks and gurgles, the frequency of which has been estimated at from five to 34 per minute. In hypothyroid patients, bowel sounds may be diminished in intensity, pitch, and frequency, and increased in hyperthyroidism.

Questions to Ask When Neck Mass is Identified:

- When was it noticed?
- What made the patient notice it?

- Any symptoms associated with the lump?
- Has the lump ever changed?
- Does the lump disappear or reappear?
- Are there other lumps in the body?
- Is there any pain with the lump?

Thyroid Enlargement Classifications

Thyroid enlargements can be classified into three main groups based on the anatomy of their enlargement, which include diffuse, multinodular, or solitary mass. Diffuse enlargement of the thyroid is defined as an enlargement of the entire gland. Diffuse enlargement of the thyroid gland may be related to hypothyroidism, iodine deficiency, Hashimoto's, Graves' disease, thyroiditis, anaplastic thyroid carcinoma, or any mechanism that promotes elevation of TSH. Multinodular enlargements, defined as multiple nodules, are basically caused by multinodular goiter. A single, solitary nodule may be due to solitary nodular goiter, colloid cyst, adenoma, thyroid carcinoma, or lymphoma.

Diffuse Enlargement

- Hypothyroidism
- Hashimoto's
- Iodine deficiency
- Graves' disease
- Thyroiditis
- Anaplastic carcinoma

Multinodular Enlargement

- Multinodular goiter

Solitary Nodule Enlargement

- Solitary nodular goiter
- Colloid cyst
- Adenoma
- Carcinoma
- Lymphoma

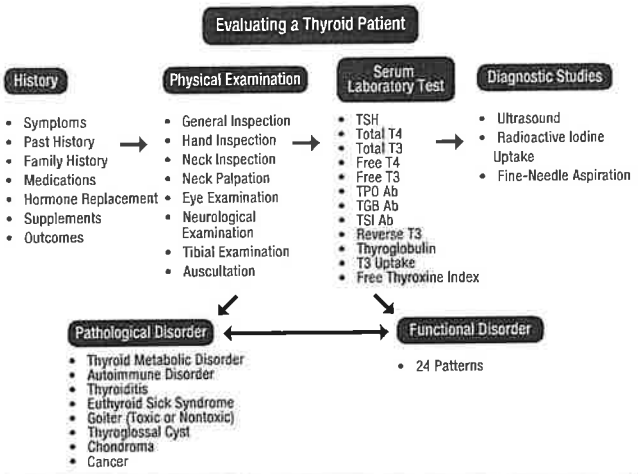
If a midline mass is noted over the thyroid, it is first important to note if the mass moves up and down with swallowing. If the mass does not move up and down with swallowing, it is most likely a chondroma or a dermoid cyst. If the mass does move with swallowing, it is then important to note if the mass moves when the tongue is protruded. If the mass moves with tongue protrusion, then it is most likely a thyroglossal cyst. If the mass does not move with tongue protrusion, then it is a thyroid enlargement and palpation of the thyroid can further subdivide the mass into diffuse, multinodular, or solitary nodule.

Chondroma: a benign cartilaginous tumor that is encapsulated with the lobe with diffuse wing patterns of cartilage. It presents as a mass over the thyroid and does not move with swallowing.

Dermoid Cyst: a benign cystic teratoma that presents as a mass over the thyroid and does not move with swallowing.

Thyroid Evaluation

Proper evaluation of the thyroid includes a detailed history, physical examination, evaluation of drug-thyroid interactions, serum laboratory tests, ultrasound, radioactive iodine uptake, and FNA biopsy. An understanding of the applications of these tests will allow the clinician to properly evaluate the thyroid condition. It is unlikely that all tests will be needed at any given time to evaluate the patient. However, the proper selection and interpretation of tests needs to be clearly understood by the practicing clinician. In this section of the manual, the utilization, interpretation, and clinical application of laboratory and diagnostic testing will be reviewed. This chapter is divided into two sections. Section I covers thyroid serum laboratory markers, and Section II covers thyroid diagnostic tests, such as ultrasound, radioactive iodine uptake, and biopsy.



Section I – Serum Laboratory Tests

Serum laboratory tests for the thyroid are the most effective way to assess metabolic defects. One area of clinical nuisance is the fact that there is no set of thyroid serum laboratory ranges that are accepted by everyone. The ranges used by the Endocrine Society, the American Thyroid Association, the American Academy of Family Physicians, etc, are all varied. Also, there is no set agreement between laboratories across the country on an established range. Laboratories typically perform a bell-curve analysis over a region at a determined time period to establish a range. This practice may not truly reflect thyroid health since thyroid status can vary across populations. Lastly, most people who have thyroid markers conducted are already taking thyroid hormones or other medications that may skew the population sample being used for the laboratory range bell-curve analysis.

In this section, you will notice a summary of all the major markers used to assess thyroid metabolism and two separate ranges. A typical laboratory range will be listed because there is no consistency between laboratories nationally. A functional range will also be listed, which is a narrower and more functional range for each marker. The functional range may be useful to identify imbalances earlier and provide a higher standard for thyroid function.

Thyroid-Stimulating Hormone (TSH)

Thyroid-stimulating hormone is also called thyrotropin. This hormone is released by the pituitary after the hypothalamus releases TRH. This is the most common marker used to assess thyroid function and is also the most sensitive. The TSH levels increase when the T4 levels drop, and the TSH falls when T4 levels increase. This is the only test performed in the traditional healthcare model as a means to screen the patient for thyroid disorders; this is because it is only concerned for screening the thyroid for hormone replacement and not optimal physiological function. A TSH test alone does not consider thyroid pituitary feedback loops, peripheral thyroid metabolism, or potential or active risk factors as identified by antibody testing. A high TSH with or without changes in T4 or T3 is diagnostic for hypothyroidism. If the thyroid is not making enough T4, the pituitary will pump out TSH to stimulate its production. A low TSH is used to determine hyperthyroid activity. If the thyroid is overactive, such as in Graves' disease, the antibodies bind to active thyrotropin receptors on the thyroid cells and stimulate T4 production without the influence of TSH. Please note that some antibodies may inhibit thyroid function by inactivating instead of stimulating thyrotropin receptors. This is called an autoimmune hypothyroid. These patterns demonstrate a hypothyroid pattern (elevated TSH) with elevated thyroid antibodies.

TSH Ranges	
Typical Laboratory Range‡	0.5-5.5 mIU/L
Functional Range	1.8-3.0 mIU/L

‡Pathological lab results will vary from region to region based on the population characteristics.

Pathological Range Indications

TSH High In:

- Primary hypothyroidism
- Hashimoto's chronic thyroiditisa
- Thyrotoxicosis due to pituitary tumor
- Thyrotropin-producing tumor
- Hypothyroid patients receiving insufficient thyroid replacement hormone

High TSH
For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- TSH levels

TSH Low In:

- Primary hyperthyroidism
- Treated Graves' disease
- Secondary and tertiary hypothyroidism
- Euthyroid sick disease
- Overdose of thyroid hormone in treatment of hypothyroidism

Functional Range Indications:

Thyroid-stimulating hormone is increased in primary thyroid hypofunction. Thyroid-stimulating hormone is decreased in thyroid hypofunction secondary to anterior pituitary hypofunction and in thyroid hyperfunction

Low TSH
For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- TSH levels

Total Thyroxine (Total T4)

The TT4 test measures both bound and unbound thyroxine levels; therefore, it does not give the activity of T4 when measured alone. This test is best completed with a T3 uptake. The free thyroxine index (free T4) can be calculated by using the T3 uptake, and demonstrate a level of T4 activity. Total T4 levels can be altered by many drugs (see Category of Medications That Impact Thyroid Physiology).

Total T4 Ranges	
Typical Laboratory Range‡	5.4-11.5 µg/dL
Functional Range:	6-12 µg/dL

Pathological Range Indications

Functional Range Indications:

The total T4 is increased with depressed TSH in thyroid hyperfunction. The total T4 is decreased with elevated TSH in thyroid hypofunction. The total T4 is decreased with low TSH in a functional pituitary hypofunction.

Free Thyroxine Index (FTI)

As stated earlier, the total thyroxine and the T3 uptake must be used together to calculate the free T4. The index is measured by multiplying the total T4 levels by the T3 uptake levels. The result is the free T4, and it determines the amount of active T4 available. Prescription drugs, as will be discussed, will always impact T4 and resin T3 uptake levels in opposite directions due to its impact on binding sites. If the total T4 level is depressed, then the T3 uptake is high; if the total T4 is elevated, the resin uptake is low. Please note that even if you are taking drugs that may impact thyroid binding, the free thyroxine index should be within the normal range if your thyroid is functioning normally.

Free Thyroxine Index Ranges	
Typical Laboratory Range‡	4.6-10.9 mg/dL
Functional Range	1.2-4.9 mg/dL

Pathological Range Indications

Free Thyroxine Index High In:

- Can be used to help identify hyperthyroidism or metabolic conditions related to decreased thyroid binding levels in association with other laboratory markers. It is not a reliable marker in isolation.

High Free Thyroxine Index
For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- Free thyroxine index levels

‡Pathological lab results will vary from region to region based on the population characteristics.

Free Thyroxine Index Low In:

- Can be used to help identify hypothyroidism or metabolic conditions related to increased thyroid binding levels in association with other laboratory markers. It is not a reliable marker in isolation.

Low Free Thyroxine Index For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- Free thyroxine index levels

Functional Range Indications:

The FTI is increased in thyroid hyperfunction. The FTI is decreased in thyroid hypofunction and with a selenium need.

Free Thyroxine (Free T4)

The free thyroxine test is used to measure the amount of free or active T4 in the blood. All the factors such as drugs and physical conditions that may impact the total T4 do not impact the free T4. The level of T4 in the blood is high with hyperthyroidism or low with hypothyroidism. Please note that even a high TSH with normal T4 is enough to diagnose hypothyroidism. A rare pattern is an elevated T4 without hyperthyroidism, which may be related to a hereditary condition of thyroid resistance. Elevated free T4 may also be caused by patients taking heparin, or by an acute illness that may briefly cause the binding protein levels to suddenly fall. If an illness becomes severe and chronic, it may decrease the free T4 levels but it is not a thyroid disease.

Free T4 Ranges	
Typical Laboratory Range‡	0.7-1.53 ng/dL
Functional Range	1.0-1.5 ng/dL

Pathological Range Indications

Free T4 High In:

- Hyperthyroidism
- Disorders of decreased thyroid-binding globulins
- Overdose of thyroid hormones

Free T4 Low In:

- Hypothyroidism
- Disorders of increased thyroid-binding globulins
- Insufficient thyroid hormone replacement
  Hypothyroidism treated with T3 replacement

Total Triiodothyronine (Total T3)

Total T3 is a quantitative determination of the total T3 concentration in the blood and is the test of choice in the diagnosis of T3 thyrotoxicosis. Total T3 is not the same test as the T3 uptake test that measures the unsaturated thyroid binding concentration in serum. The test is used to diagnose hyperthyroidism. Total T3 may be useful to identify peripheral down regulations in the enzyme 5'-deiodinase, which converts T4 into T3.

Total T3 Ranges	
Typical Laboratory Range‡	60-181 ng/dL
Functional Range:	100-180 ng/dL

Pathological Range Indications

Total T3 High In:

- Hyperthyroidism
- T3 thyrotoxicosis
- Daily dosage of >26 ng/dL of T3
- Acute thyroiditis
- TBG elevation for any cause
- Daily dosage of >300 µg/dL of T4

Total T3 Low In:

- Hypothyroidism; however, some clinically hypothyroid patients will have normal levels
- Starvation
- Down regulation of 5'-deiodinase
  - Elevated cortisol
  - Increased lipid peroxidation

Resin T3 Uptake

The resin T3 uptake measures the amount of sites for active (unbound) T3 to bind on thyroxine-binding proteins. This test is performed by mixing the blood with radioactive thyroid hormones. These radioactive hormones then combine with binding sites on thyroxine-binding proteins. The blood is then exposed to a

‡Pathological lab results will vary from region to region based on the population characteristics.

substance called a resin that will bind the unbound thyroid hormones and measure for radioactivity. The result can be expressed as the percent of radioactivity found in the resin, compared to the original radioactivity that was added. The more binding sites that are open on the proteins, the lower the resin uptake result will be, and vice versa. For example, anything that reduces the binding sites, such as elevated testosterone or testosterone replacement therapy, causes a low T4 measurement because it leaves very few binding sites for any more thyroid hormone to bind to it. If T3 is added to the sample of the blood, little T3 will be bound. This pattern would have low total T4 levels and high resin T3 uptake levels. On the other hand, anything that raises the binding sites, such as estrogen or birth control pills, would cause a pattern of high total T4 and low T3 uptake.

Resin T3 Uptake Ranges	
Typical Laboratory Range‡	24%-39%
Functional Range	28%-38%

Pathological Range Indications

T3 Uptake High In:

- Medications
  - Heparin
  - Anabolic steroids (testosterone)
  - Phenytoin
  - Dicumarol
  - Large doses of salicylates

High T3
For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- T3 levels

T3 Uptake Low In:

- Normal pregnancy
- Medications
  - Heparin
  - Estrogens
  - Methadone
  - Anti-ovulatory drugs

Low T3
For treatment of this condition:
Seek appropriate conventional medical management and drug administration by a physician.

Consider diet, nutrition, and lifestyle support for:

- T3 levels

Functional Range Indications:

T3 uptake is increased with thyroid hyperfunction and with testosterone elevation. T3 uptake is decreased with thyroid hypofunction and with estrogen elevation.

Free Triiodothyronine (Free T3)

This test measures the free T3 hormone levels. This test is rarely completed in traditional endocrinology. It is typically only used in a situation when a patient has hyperthyroid symptoms, yet the free T4 levels are normal. However, the free T3 test is the best marker to see what amount of active thyroid hormones are available for the thyroid receptor sites.

Free T3 Ranges	
Typical Laboratory Range‡	2.3-4.2 pg/mL
Functional Range	3.0-4.0 pg/mL

Pathological Range Indications

Free T3 High In:

- Hyperthyroidism
- Thyrotoxicosis
- Peripheral resistance syndromes

Free T3 Low In:

- Hypothyroidism
- Third trimester of pregnancy
- High cortisol conditions

Reverse T3

This test measures the amount of reverse T3 that is produced. The production of reverse T3 typically takes place in cases of extreme stress, such as major trauma, surgery, or severe chronic stress. It appears that the increased production of reverse T3 is due to an inability to clear reverse T3, as well as from elevated cortisol.

‡Pathological lab results will vary from region to region based on the population characteristics.

Reverse T3 Ranges	
Typical Laboratory Range‡	9.2-24.1 ng/dL
Functional Range	9.2-24.1 ng/dL

Pathological Range Indications

Reverse T3 High In:

- Elevated stress response
- Elevated cortisol

Reverse T3 Low In:

- Severe tissue injury (burn) and/or inflammation
- Liver disease or cirrhosis

Thyroid-Binding Globulin

"Thyroid-binding globulin (TBG) binds thyroid hormones in circulation. It is the main protein of the three proteins (along with transthyretin and albumin) responsible for carrying both T4 and T3 in the bloodstream." The test may help determine patterns of thyroid imbalances from other factors such as drugs, liver disease, etc.

TBG Ranges	
Typical Laboratory Range‡	15-30 µg/mL
Functional Range:	18-27 µg/mL

TBG High In:

- Pregnancy
- Active hepatitis
- Biliary cirrhosis
- Infection
- Acute intermittent porphyria
- Familial hyperthyroxinemia
- Drugs
  - Oral contraceptives
  - Estrogens
  - Tamoxifen
  - Methadone—used for the treatment of narcotic withdrawal and dependence
  - Heroin
  - Perphenazine—used as an antipsychotic

TBG Low In:

- Chronic liver disease
- Obstructive jaundice
- Liver cirrhosis
- Severe systemic illness
- Active acromegaly
- Nephrotic syndrome
- Surgical stress
- Medications
  - Androgens
  - Salicylates
  - L-asparaginase—used for leukemia and mast cell tumors
  - Large doses of glucocorticoids (prednisone)

Thyroid Antibodies

Thyroid autoantibodies indicate that the body's immune system is attacking itself. Production of thyroid autoantibodies may create a hypothyroid or a hyperthyroid state. They may also be elevated in cases of thyroid malignancy. Some antibodies attach to the TSH receptors but do not cause a response; therefore, the patient will complain of low thyroid symptoms; however, the serum TSH may not be altered. It is just not able to cause a cellular change. On the other hand, some antibodies will bind to the receptor sites and cause overactivation of the thyroid. This will present as elevated T4 levels, a low TSH, and elevated thyroid antibodies. There are three thyroid antibody tests:

- **Thyroid Peroxidase Antibody (TPO Ab):** This is the most commonly elevated antibody in autoimmune thyroid responses. TPO Ab is also known as microsomal antibody.
- **Thyroglobulin Antibodies (Tg Ab):** This is not as commonly elevated with autoimmune thyroid conditions as TPO Ab. It is typically ordered when thyroid lab results seem awkward since the antibody responses may interfere with thyroid hormone production. Tg Ab testing is also typically conducted in regular intervals with thyroid cancer treatment.
- **Thyroid-Stimulating Hormone Receptor Antibody (TSH-R Ab):** This antibody is only ordered when the patient is hyperthyroid. Positive results strongly suggest Graves' disease. This test is also known as thyroid-stimulating immunoglobulin (TSI).

‡Pathological lab results will vary from region to region based on the population characteristics.

Section II – Thyroid Diagnostic Testing

Thyroid Ultrasound

Thyroid ultrasound is the least invasive way to evaluate a solitary nodule in the thyroid. It simply involves having the patient lie comfortably with transmission gel on his or her throat as a sound-emitting instrument is gently massaged across the thyroid gland. How the emitted sound bounces back to the instrument can provide information to determine if the nodule has characteristics of a benign or malignant tissue. If the nodule has benign characteristics, a follow-up may be conducted in six months to a year to evaluate the gland again. If the nodule has malignant characteristics, a FNA biopsy or radioactive iodine uptake (RAIU) test may be conducted.

Malignant Characteristics of Thyroid Ultrasound

- Irregular border
- Hypoechoic (less echogenic than the surrounding tissue)
- Microcalcifications
- Taller-than-wide shape on transverse study
- Significant intranodular blood flow by power Doppler

Benign Characteristics of Thyroid Ultrasound

- Smooth borders
- Hyperechoic
- "Comet tail" artifact as sound waves bounce off intranodular colloid

Radioactive Iodine Uptake (RAIU)

An RAIU scan is a test conducted to evaluate the metabolic activity of iodine uptake by the thyroid gland. The test involves either oral ingestion of radioactive iodine or intravenous injection, followed by a radioiodine scan. "Most often, scanning is done 24 hours later if the radioisotope is given orally. If it is given intravenously, the scan is performed approximately 20 minutes later. For a thyroid scan, the patient is positioned lying down on his or her back, with the head tilted back. The radionuclide scanner, also called a gamma camera, is positioned above the thyroid area as it scans. The scanning takes 30-60 minutes."

"A normal scan will show a thyroid of normal size, shape, and position. The amount of radionuclide uptake by the thyroid will be normal according to established laboratory figures. In a normal scan there will be no areas where radionuclide uptake is increased or decreased. An area of increased radionuclide uptake may be called a hot nodule, or 'hot spot.' This means that the tissue is overactive." If there is a diffuse hot spot, it indicates that the entire gland is overactive, which is typically found with Graves' disease. If there is a localized hot spot, it indicates the presence of a toxic adenoma, which may need to be treated with surgery. Areas of "decreased radionuclide uptake may be called a cold nodule, or 'cold spot.' This indicates either" an area of decreased uptake that is not active in utilizing iodine, or that is not thyroid tissue. A localized cold spot strongly suggests the possibility of thyroid cancer. Cancer cells do not take up iodine. If the entire thyroid gland demonstrates low radioactive iodine uptake, it suggests thyroiditis. The RAIU test is typically not used to establish a clear diagnosis as an isolated test. It must be used with thyroid lab tests. It is typically a test conducted with hyperthyroid patients to help differentiate if the mechanism for thyroid hyperfunction is Graves' disease, toxic adenoma, or thyroiditis. The thyroid-stimulating immunoglobulins (TSI) test is currently the standard test to rule in or out the diagnosis of Graves' disease and RAIU. The RAIU test is commonly used in hyperthyroidism if the diagnosis of Graves' cannot be made due to negative TSI results. Listed below is the summary of potential diagnoses with uptakes that are increased, decreased, localized, and/or diffuse:

Diffusely High RAIU With Hyperthyroidism

- Thyrotoxicosis (Graves' disease)

Localized High RAIU With Hyperthyroidism

- Hyperfunctioning thyroid nodule (toxic adenoma)

Diffusely Low RAIU With Hyperthyroidism

- Subacute thyroiditis
- Postpartum thyroiditis (5%-8% of incidence)
- Painless lymphocytic thyroiditis
- Medications
  - Exogenous thyroid hormone intake
  - Excessive exogenous iodine intake
  - Goitrogenic medications (see previous list under "Goiter")